COMPLETE USER GUIDE

UO Asset Studio

From your first profile to a finished release.

Version 0.33.13 | Windows desktop edition

Simple / Advanced | Hover help | Native VD conversion

Created by Celobra

Find | Prepare | Preview | Review | Save | Verify

33 chapters covering every Studio workspace and tool.

Step-by-step workflows, controls, recovery and troubleshooting.

Offline documentation. Generic examples; no personal profiles included.

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Using this guide

Created by Celobra.

Complete user guide | Version 0.33.13 | Windows desktop edition

Browse, create, test and distribute UO content, with new-script generators for Sphere, RunUO, ServUO and ModernUO.

This manual describes the 0.33.13 interface. It covers the main views, importers, workshops, House Builder, World Generator, file tools and recovery workflows. Button names match the application; a trailing ... or … means that a button opens another window. Examples use generic names and do not identify free IDs on your installation.

The HTML edition works offline and includes a searchable contents panel. The PDF has a linked contents page and bookmarks. Keep the Markdown edition if you want to adapt the documentation for your own distribution. No game files, accounts, profile settings or private test projects are included.

01. Start here

What Studio does

UO Asset Studio connects your server profile to your Ultima Online client assets. Sphere profiles provide definitions, references and script editing; RunUO, ServUO and ModernUO profiles provide read-only C# source browsing and target-specific new-script generation. You can import replacement or new artwork, preview it, and review the files that will change before saving. Artwork-only imports do not require generating a server script.

Studio is a native desktop application. Normal editing does not require a browser or a running server. Its Test Scene and House Builder walk test run inside the application. They show visual appearance and local movement; they do not execute your Sphere scripts.

The four things to keep separate

Thing	What it contains	How you preserve it
Working game files	Client art, animations, metadata and server scripts	A reviewed file save; a retained backup allows restoration
Editing session	Script drafts, open script-workspace documents and reviewed pending changes	Automatic checkpoints, or Asset workspace > Checkpoint now
Editable project	A house design, gump design or animation sequence	That workshop's Save project, Save design or Export edited sequence
Distribution	The content or documentation you intend to share	A deliberately prepared pack, export or distribution folder

A session checkpoint does not save game files. A screenshot does not save an editable design. A profile does not make an independent copy of your client or server.

Start with animation conversion alone

Studio opens Home even before you link client or server folders. Choose Animation workshop & VD converter to turn PNG/BMP frames, a GIF or a supported exported body folder into a VD file. This file conversion works independently of your game installation. Chapter 16 explains layout selection, stored directions and incomplete action coverage.

Use Choose project folders when you want to browse linked content or import changes into client/server files. The first editing-session steps below describe that setup.

A first session in ten steps

1. Install Studio, then open it from the Start menu.
2. Choose the server family and link its server root, Scripts or ModernUO UOContent folder, plus the matching UO client-data folder.
3. Give the profile a clear name, such as `Development shard`.

4. If you need a separate working copy, use All tools > Create test copy and select the resulting profile.
5. Keep Simple selected. For Sphere, press Ctrl+K, open Browse definitions and search for a familiar item. For C#, open Scripts and select a file in Script files.
6. Inspect the selected Sphere definition, Scripts and Item artwork, or read the selected C# source without editing it.
7. Open Test Scene to see the asset against a floor and walls.
8. Make a small change using the appropriate editor or importer.
9. Read the save review, including destination IDs, file formats, scripts and backup choice.
10. Save, then verify the result with the client and server that will actually use the files.

For a first experiment, copy existing equipment or add an item that reuses existing artwork. A new creature, wearable or house has more connected parts.

Reading this manual

Each chapter explains where a tool lives, how to use it and what is saved. Chapters 3 and 4 explain saving and recovery and apply throughout the manual. The task finder near the end helps you choose a tool. Troubleshooting covers common errors without asking you to start over blindly.

Recent additions

Version 0.33.13 adds Sphere, RunUO, ServUO and ModernUO profile setup and explicit Artwork only imports. Choose Server type and the actual server/script/content location; C# source opens read-only in Script files. Item, animated-item, visual-effect and ordinary body imports start without generated scripts. Choose Artwork and server script above the tabs only when you need matching source. Chapters 2, 3 and 10-12 explain the choices and the complete mount/equipment workflows.

Version 0.33.12 keeps playback and image export together in Animations. Choose All animations, then Body animations or Animated items & effects, search an ID or name and double-click a result. Play/Pause, stepping, the frame slider, Save GIF and Export frame work in the same viewer for both kinds of sequence. Native AnimData artwork no longer switches tabs to start playback; Preview in Test Scene is an optional scene preview. Body sources retain their raw bank/group identity, and raw UOP previews and routed VD conversion remain labelled separately. Chapters 3, 15, 21 and 29 explain the steps and source notices.

Version 0.33.10 makes graphic destinations and intentional replacement clearer for statics/tiles, gumps, animation bodies and mounts. Asset spaces labels Free, Used, Referenced, Queued and Unavailable destinations and explains their reasons. A supported replacement needs a fresh confirmation in the final save review, with backups on by default for that replacement. Chapters 4 and 8 explain these checks and their limits.

Version 0.33.8 improves static folder and ZIP imports. Compatible tiles load together while unusable images are reported by filename and reason. Static pack batches support up to 4096 compatible tiles within the existing size budgets, and PNG preference recognises exact mirrored format folders. Chapter 10 explains the source choices, View skipped files report and separate animation limits.

Version 0.33.7 improves Map Regions navigation. Pan with canvas-focused arrows or mouse dragging, double-click a region name to locate it, and fit or zoom the view before editing. Chapter 33 explains the controls and terrain-load safeguards; chapter 30 lists the keyboard and mouse actions.

Version 0.33.6 improves World Generator with coherent landscape presets, kept/comparable worlds, selective variations, queued biome painting, setup favourites, live phase pictures, named destination plots and routes, catchment-based water, scenery themes, building stamps and varied interiors. Native terrain review adds transition and route diagnostics. The application audit improves smaller-window navigation and script-workspace

search, scrolling, toolbar layout, symbol refresh and UTF-8 review. Portable Sphere packs preserve their text encoding, recheck duplicate declarations and filenames against existing and queued scripts, and create reviewed nested folders transactionally. Chapters 3, 5, 28 and 32 explain the changes.

Version 0.33.5 adds reusable New script, Add to existing script and staged New folder controls to asset imports and script generation. Review the complete resulting Sphere file with its original definitions/comments before saving or queuing. Whole armor sets can use an existing file with One file for whole set; C# retains new separate files. Chapters 10, 14 and 28 explain destinations and distribution.

Earlier improvements are covered in their relevant tool chapters. For the full release history, read CHANGELOG.md in the offline guide bundle, installed documentation or source download.

02. Installation, updates and profiles

Install or upgrade

Run the supplied Setup executable. The installed app includes its runtime; recipients do not need to install Python, Pillow, Inno Setup or developer tools.

To upgrade, save your work and close Studio, then run the new Setup. It detects the earlier installation and reuses its installation location. The confirmation step offers Upgrade, or Repair when installing the same version. Uninstalling first is normally unnecessary.

The default installation location is `%LOCALAPPDATA%\Programs\UO Asset Studio`. An older installation under Program Files may need Windows administrator permission to replace its application files. Use the same Windows account for that upgrade. The installer does not replace your linked shard/client data or saved profiles.

Keep `UOAssetStudio.exe` with its `runtime` folder. The `docs` folder contains documentation; `licenses` contains dependency notices. Do not keep your artwork projects inside `runtime`.

Check for a newer version

Starting with 0.32.3, Studio checks the Studio website in the background when it opens. When the website lists a newer version, an Updates window offers the available release. The offer waits while a reviewed save or a modal dialog is open.

1. Choose **Download & install** to download the newer Windows installer, or **Later** to continue working.
2. Follow the download progress. **Cancel download** stops the download and keeps Studio open.
3. Studio verifies the complete download against the website's file size and SHA-256 checksum. A failed or interrupted download is not run.
4. Finish open dialogs and saves, then resolve the normal prompts for script drafts and pending changes. Cancelling an exit prompt keeps Studio open; it does not schedule installation for your next ordinary exit.
5. Studio closes and the usual Setup window opens. Review **Upgrade** and complete the installation. Open Studio again when Setup finishes.

Open **Updates** near the bottom of the sidebar, or find it in **All tools**, to see your installed version, use **Check now**, or read the release notes. Clear **Check for updates when Studio starts** to disable startup checks; the choice applies across profiles and is remembered even before you choose a project. Manual checks remain available.

An unavailable website does not block editing. Automatic checks fail quietly; a manual check explains the connection problem and allows another attempt. Updating the application preserves your profiles and linked client/server files. The updater sends a version-check request to the website and downloads an installer only after you approve; it does not upload your project or game files.

Install 0.32.3 manually once if your older Studio has no Updates button. Running from Python source opens the website download page instead of updating that source environment. The downloaded installer uses normal Windows setup; older protected installation folders may still require administrator permission.

First-run project setup

Home is available with no linked project folders. For direct VD conversion, open Animation workshop & VD converter and select your source files. To enable client/server browsing and editing, choose Choose project folders and provide the locations below. Cancelling setup returns to Home.

Choose **Server type**: Auto, Sphere, RunUO, ServUO or ModernUO. Then use **Server / scripts** to select the server root, its Scripts folder, or a ModernUO Projects/UOContent directory. ModernUO source under Items, Mobiles and other content directories does not need to be moved into a new Scripts folder. Supported source ZIPs can be browsed without extracting them; extract the source before writing into that server.

Auto recognises supported layouts and markers. If C# is found but the fork cannot be identified, choose RunUO, ServUO or ModernUO explicitly. These servers use different APIs, so choose the actual family rather than guessing from the `.cs` extension. The profile remembers that choice. A leftover `.scp` does not turn an explicitly selected C# profile into Sphere.

C# source files appear in **Script files** for reading. They do not become Sphere ITEMDEF/CHARDEF definitions, and the Sphere definition editor, existing-script editing and region tools are not C# editors. New generated scripts use the profile's target and an extracted C# server/export folder. Switching profiles resets the generation target and C# export location so output does not silently follow the previous shard. A ZIP is not a writable output folder; choose an extracted server or separate existing export directory when generating source.

Choose the UO data folder containing the actual client files, such as TileData, art and animation archives. The optional ClassicUO executable path is a remembered location; it is not required for Studio's in-app previews. Saving a profile does not configure a running game client to use those data files.

Click Save and Open Editor. The definition list comes from the selected server; asset previews come from the selected client folder. Use Item artwork > Check game files if you suspect that Sphere is reading a different folder.

Profile controls

Control	Use
Project selector	Switch to an existing profile
Manage > New profile...	Name a profile and select its locations
Manage > Edit paths...	Correct or change the active profile's linked locations
Manage > Rename...	Change the profile's display name
Manage > Save as new profile...	Copy the profile settings under another name
Manage > Remove profile...	Remove the saved profile entry; it does not delete the linked shard
Manage > Linked folders...	See the active locations and open their folders

Two profiles pointing to the same client folder edit the same client files. Use Create test copy, or prepare separate folders yourself, for independence. The last active profile opens next time. Finish open dialogs and saves before switching profiles; resolve any prompt about drafts or pending changes.

Settings and moving computers

Profiles are normally stored in `%APPDATA%\UOAssetStudio\project.json`. Back up this file if you want to preserve the profile list. On another computer, use Edit paths to update drive letters and locations. Settings are retained across application upgrades and uninstalling.

Advanced users can launch with `--config "D:\StudioSettings\project.json"` to use another settings file. That is optional; normal recipients can use the default. If settings are corrupt or another Studio instance has saved a newer copy, follow the reported conflict message instead of overwriting the file manually.

03. The main window and daily navigation

Home and the navigation sidebar

Home opens with your project status and five artwork buttons: Import Tiles, Import Statics, Import Gumps, Import Animation and Import Mounts. Its Create or convert area links to Animation workshop & VD converter, House Builder, World Generator and Complete armor set. Explore all tools opens the complete 35-tool collection, including Generate server script.

Simple presents the common navigation destinations, including Home, Import artwork, All tools, Asset workspace, Scripts, Animations and Test Scene. Advanced also shows direct sidebar links for Definitions, Item artwork, Gumps, Hues and Asset spaces. The complete searchable collection includes these views in both modes. C# profiles keep their source-file browsing available without exposing Sphere editing tools. Map Regions opens its own in-app window. Appearance is at the bottom. The project selector and Manage menu stay in the top bar.

Animations opens with Animation library > All animations. Body animations reads the supported populated MUL/UOP body sources in the linked client folder; Animated items & effects reads native AnimData artwork sequences. Search an ID or name and double-click a result, or press Enter. A server definition is optional for this library. Shard definitions is a separate filter for entries with resolved shard names. See chapter 15 for raw/routed body preview and item/effect playback.

Choose an artwork import

Items, animated items, spell effects and ordinary body animations start with **Artwork only**, shown above the tabs in Simple and Advanced. It prepares the native client-file changes for review without generating `.scp` or `.cs` scripts, registering a Sphere resource, or adding body-remapping files. Choose **Artwork and server script** when you need matching source; then open **Server script** to configure its target and destination. **Review artwork** changes to **Review files and script** with that choice. For raw mount body data use Body / character animations; the complete Mount and equipment workflows still include their required definitions and appearance/riding links.

Choose a button on Home or open Import artwork in the sidebar. The Import artwork page keeps the five choices together and also links to Creature or monster, Complete armor set, Clothing or weapon, Animated items, Spell / visual effects, and the separate animation converter.

Button	Use it for	What opens
Import Tiles	Floors, walls and roofs made from item-art images	Item artwork import, including folders and ZIPs
Import Statics	Items, furniture and decorations	The same item-art import with an object context
Import Gumps	Interface and paperdoll images	Native gump-image import
Import Animation	Animated items, spell/visual effects, body animations or mounts	A choice of the appropriate animation workflow
Import Mounts	Mount animation, inventory icons and riding links	The mount animation and server/riding workflow

Tiles and Statics use the same client item-art format. Their names help you choose a starting point; they do not set collision, surface, height or other TileData flags automatically. Ground terrain belongs in World Generator rather than these two imports. Chapters 10-12 and 17 explain the source, destination and review steps.

Choose Import Animation, then choose the kind of animation:

Choice	Source and result
Animated items	GIF or ordered PNG/BMP frames for an animated object. The importer starts with one animated static enabled, consecutive free artwork IDs and native TileData/AnimData changes. Add an item script on Server script when needed.
Spell / visual effects	GIF or ordered PNG/BMP frames for fire, water, auras or other effects. The same native animated artwork is prepared; Server script optionally adds an item definition and callable visual-effect function.
Body / character animations	A compatible VD, GIF or ordered frames in a native body-animation layout. Server-script generation starts disabled. Use Creature or monster when you also want creature gameplay setup.
Mounts / rideable objects	Mount body animation, inventory icons and riding/server links. Its required actions and client mappings still need checking.

Loading files does not save them. Inspect the preview, review the proposed changes, then choose Save original files or Queue changes. GIF delays do not automatically become native item timing; inspect Frame interval for animated items/effects.

To share animated item/effect artwork, queue its reviewed changes and use Content packs > Export pending bundle before saving the queue. The .uoasset file is a ZIP container for supported art, TileData, AnimData and linked Sphere script changes. VD stores native body-animation data; it does not package item art or AnimData. Chapters 10 and 28 explain the item/effect workflow and recipient remapping.

To create a distributable body VD without installing it, choose Animation workshop & VD converter. To export an existing client body with its available directions, use the body library or the manual Animations > Source: Auto > Load body > Save entire body as .vd route. A raw UOP library selection has a separate Convert routed body to .vd command because the client layout may select different groups. Chapters 15 and 16 explain these routes and action coverage.

Choose Simple or Advanced

Use the Simple / Advanced switch above the active view. Supported importers and the animation workshop also show the switch. Changing it affects the shared application mode, and Studio remembers the preference for the project.

Mode	What you choose	What Studio handles
Simple	Source files, the kind of asset, required artwork or layout, names and gameplay properties	Supported importers select free space and use their automatic format/alignment defaults; detailed replacement controls are hidden

Mode	What you choose	What Studio handles
Advanced	The same inputs plus destination IDs, supported file formats, alignment and replacement options	Studio validates the chosen destinations and still prepares a save review

Start in Simple when you want to bring in a new asset. Choose files, inspect the preview and review the proposed change before saving. A source animation still needs the appropriate layout; an inventory icon still needs its related worn appearance where applicable. Automatic settings do not create missing artwork, directions or actions.

Switch to Advanced when deliberately updating an existing record, adjusting frame registration or choosing a specific destination. The detailed controls remain available, and returning to Advanced restores remembered manual choices in supported importers. A selected mode does not save game files by itself. Chapter 4 explains review, queues and recovery.

Read the hover explanations

Pause the mouse over a button, option or field to see its explanation. Help appears after a short delay and disappears when you move away, click or type. It describes the input, affected files or preview, and relevant limits for supported controls. Follow the visible status text and save review as well: they report the actual result and destinations for your operation.

Find a tool and keep the editor clear

Press Ctrl+K or choose Find a tool. Search by a task such as armor, sound, map or save, or choose Import, Assets, Scripts, Worlds or Files. Results show their group headings: Import artwork, Browse & preview, Server scripts, Build worlds, and Files & distribution. A filter hides groups with no matching tools. Enter opens the first matching result. Ctrl+H returns Home. Alt+Left / Right move through your view history.

Simple keeps unrelated editor side panels out of the common workflow, while Scripts keeps its Script files pane visible. In Advanced, Definitions and Asset workspace initially show the definition browser; Scripts initially shows Script files. Use the Definitions and Script files switches above an editor to choose which side panels are visible. The app remembers the choice for each view and keeps your open script text when switching views. C# files remain read-only in either mode. Press Ctrl+K to reach any view from either mode.

Drag dividers to resize the visible panels. Restored widths are checked so an editor or side panel is not left collapsed. Appearance > Reset panel layout restores the defaults. The minimum main-window size is 1100 x 720; some editors have their own minimum sizes or scrollbars.

Save All and Pending changes remain in the toolbar. Save options beside Save All contains Save scripts only, Save game files only and Save & recovery centre. Manage contains New profile, Edit paths, Linked folders, Rename, Save as new profile and Remove profile.

Search definitions

1. Enter a name, DEFNAME, identifier, path fragment or description in Search Sphere definitions.
2. Use the type selector to narrow the results to items or creatures.
3. Choose or type a Category / subsection. Empty search text browses the chosen category.
4. Select a row to populate the definition and script views.

5. Use Preview in Test Scene to open the selected content in the local scene.

All categories removes the category restriction. Categories come from the shard's actual CATEGORY and SUBSECTION fields. Definitions without those fields are still searchable. Numeric IDs and named identifiers are different ways to locate content; read the result's type before interpreting its number.

Definition and Workspace

Definition shows the selected section's properties, artwork and indexed references. Trigger assignments remain in the script; they are not all treated as top-level static properties. Double-click a reference to follow a resolved target. Back and Forward return through navigation history.

Workspace collects the current asset's related resources, save/session status and favourite/recent selections. Preview opens the relevant asset view. Find references starts the broader reference scan. Favourite toggles the current asset in your saved list. Detach preview opens its image separately. Export PNG saves the available image. Double-click related resources to navigate where supported.

Selecting a creature identifies a body animation; it does not turn that body number into an item-art ID. Use Selected creature in Animations when you want to follow a CHARDEF.

Themes

Appearance and the Theme menu offer Classic Studio, Invicta, Obsidian, Midnight Blue, Forest, Crimson, Royal Violet, Copper, Parchment and Arctic. Choose one to recolour the interface immediately. The selection is remembered per project. Appearance also offers Standard or Larger text and Reset panel layout. Themes do not recolour the original artwork or replace Test Scene's lighting controls.

Shortcuts to common tools

All tools provides the complete searchable 35-tool collection under Import artwork, Browse & preview, Server scripts, Build worlds, and Files & distribution. The category selector uses the shorter names Import, Assets, Scripts, Worlds and Files. Earlier searches such as Static tiles from folder / ZIP and Mount or rideable object find Import Tiles and Import Mounts. Use the collection for tools that previously appeared as separate toolbar buttons or repeated menus. Right-click menus provide actions for the item, gump, animation or script you clicked. Select the intended asset before using an action based on the active selection.

Select a definition, then use Asset workspace > Sound in UO > Attach sound to this asset... to link audio to the displayed item or creature. This panel sits above the large preview, so its button is immediately visible. It follows the asset shown in Workspace even if you later open another script. If you browse a raw asset without a linked definition, select its item or creature definition first. Right-click > Attach sound... remains available in the definitions list. For an effect function, open its script and use All tools > Attach a sound. Chapter 20 explains the sound controls.

Drag-and-drop supports a project script, an exported animation body folder, or image files. A single image dropped over an active item/gump can open replacement of that ID. An image batch offers item/static import or animation-frame import. A body folder opens the animation workshop for destination setup. A dropped script must belong to the active project. Use the normal file buttons when drag-and-drop is unavailable.

Recent activity

Open **Recent activity** from the sidebar in either mode to see the latest 100 completed Studio actions across your projects. Imports and script saves appear after their files have been written; exports and saved workshop projects also appear. Previews, cancelled choices and failed saves do not count as completed actions.

The newest action appears first, with its time, task and project. Select a row to read its details and affected files. Search narrows the displayed list; Refresh rereads it. Open affected folder opens an available folder for the selected action. The list survives restarting Studio and stays on this computer; it does not send activity to the website or act as an undo history. Use the save/recovery tools for restoring files.

04. Saving, queues, backups and recovery

Choose the correct save command

Command	What it does
Validate and Save in Scripts (Sphere profiles)	Writes the displayed section or opened script document to its original .scp file
Save scripts	Saves the current script, or reviews changed script-workspace documents when applicable
Save game files	Saves the reviewed pending operations
Save All	Combines pending operations with changed script text and script-workspace tabs
Queue changes in a review	Adds that reviewed snapshot to Pending changes; does not yet write game files
Save original files in a review	Writes that reviewed operation immediately
Save project / Save design	Saves the workshop's editable document, separate from game files

Before writing client/server files, close the game, Sphere and other editors using those files. Reopen or reload the relevant client/server after saving so it reads the changes. Studio saving a script does not execute that script on a running server.

Automatic checks before saving

For a reviewed game-file save, Studio first checks every destination and the other native client archives and mappings in the selected client-data folder. On Windows, it tests exclusive access to those files and the permissions needed to write the destinations and create save files in their folders. Even an open reader that normally permits shared writing can stop the check.

All initial checks finish before Studio creates its save lock, backup, recovery journal or staged output. If a file is locked, missing unexpectedly, read-only or inaccessible, the Save safety check lists the affected paths and stops the operation. An initial failed check makes no backup or partial game-file change.

1. Read the reported filenames and reason for each failure.
2. Close the game, Sphere or editor using those files.
3. If access is denied, check that the active profile points to the intended writable working folder and that your Windows account has access.
4. Retry the save. If another editor changed the files, refresh and review your changes again when prompted.

The file-access check repeats after output preparation, immediately before the first replacement. Direct Sphere scripts saves check the target script before making a backup, and again before replacing it. Recovery and restore check their destinations before restoring any file. Existing conflict checks and recovery protection still apply.

Keep the relevant programs closed until saving finishes. A file-access check cannot detect an application that has already read its data and closed its file handles, and another program could open a file after the check. Studio does not close applications or change permissions for you. These checks do not make a multi-file save indivisible or replace retained backups.

Read a save review

For graphic imports, choose **View graphics review details…** to inspect the affected source and bank, destination IDs and ownership reasons. Body details distinguish raw native body IDs from linked logical server IDs. The list also explains queued, invalid or unsupported-override destinations. Choosing an importer replacement option or selecting a manual destination only prepares a proposal.

If existing graphics are affected, the checkbox **I confirm replacing the existing graphics listed in this review** starts unchecked. Inspect the listed content, then confirm only the exact replacement you intend. Save and Queue stay disabled until the required confirmation is given. A queued or unavailable destination cannot be made writable by ticking this box. Choose other space or reconcile the existing pending operation.

The confirmation belongs to the exact proposal and checked files. Changed source files, destination IDs, mappings, operation contents or pending reservations require a fresh review. Restoring a session also requires fresh replacement consent; a saved session does not approve a later overwrite.

Read the complete list of IDs and affected files. Where present, compare original and converted artwork, listen to converted audio, inspect frame coverage, and read the generated script. Check the chosen MUL/UOP destination and any linked mapping or resource-table changes. Back returns to the originating editor without saving.

Keep a backup is optional for ordinary saves and normally off. Reviews that replace existing graphics start with Keep a backup enabled. Retain originals before deliberately replacing content that other definitions may share. Removal operations keep a backup automatically. A save journal used to recover an interrupted operation is not a substitute for deliberately retaining backups of completed saves.

Follow save progress

Reviewed game-file saves and Save All show a moving activity bar, the current phase, the filename and its position in the reviewed bundle, plus elapsed minutes and seconds. The bar shows continuing activity; it is not an estimate of the time remaining. Large archive or resource-table saves can take longer than small script edits.

After writing, the status says Files saved. Refreshing and shows script-reading counts, reference resolution and folder refresh. Studio does this work in the background once per save; binary-only changes skip unnecessary script indexing. Close, Back and save controls remain disabled until the operation and refresh finish. There is no cancellation midway through a game-file save.

If refresh fails after writing, Studio reports that the files were already saved. The completed queue is cleared; refresh the project before reviewing further changes. Open unsaved script text is retained.

Work with the queue

Open Pending changes to inspect queued operations. Remove selected drops an operation from the queue; it does not undo an already completed file save. Use the backup checkbox in the pending-save window to retain originals for the combined save.

The queue holds a snapshot. Changing an import form or script after queuing it does not update that snapshot. If the new draft conflicts with the queued one, remove the old queue entry, review the current draft and queue it again. Identical script drafts can be recognised by Save All; different edits to the same binary record are not merged by guessing.

Separate operations can share an archive when they change different records. Automatic allocation considers supported pending imports, so their destination IDs are reserved during subsequent allocation. This is a local planning aid, not a reservation across another program or another person's edits. Destinations are checked again when saving.

Script backups and manual backups

For Sphere profiles, Scripts has Back up original and an optional Also create a backup when saving checkbox. The script directory also offers Back up before opening for edit. These controls concern Sphere script-file backups. Back up original copies the on-disk file, not unsaved text in the editor. Extracting a C# project does not enable these editing or backup controls.

Reviewed client-file backups live in `.uopve-save-...` folders beneath the client-data folder. Script-editor backups use `.uopve-backups`. Keep a reviewed-save backup folder intact, including its journal and linked originals.

Recover or restore files

1. Open Asset workspace > Save / recovery centre, or Item artwork > Recover client save.
2. For a failed or interrupted operation, choose Recover interrupted save.
3. For an intentional rollback, select a retained completed backup and choose Restore selected backup.
4. Save or discard current drafts and pending edits as directed, and close programs using the files.
5. Read the operation details and confirm restoration.
6. Reload the project and inspect the restored content.

Restoration applies to the complete linked operation: client assets, scripts and mappings. A new script created by that operation can be removed during restoration. If a file has changed again since the backed-up save, restoration can stop to protect the later work. Do not force an old entire archive over newer content; resolve which changes to retain first.

Restore an editing session

Asset workspace > Checkpoint now records drafts and the reviewed queue. Normal editing also schedules checkpoints. After reopening the same project, Restore session can recover them. The recovery centre also offers Open archived session and Discard saved session. Discarding a stored session does not erase game files or original-file backups.

Session queues depend on the files they were reviewed against. When those files change, Studio can retain draft text while requiring imports to be reviewed again. Sessions belong to the relevant project locations; they are not portable content packs.

Save unfinished animation edits with Export edited sequence and unfinished gump layouts with Save design. House Builder has its own Save project and automatic recovery. Do not assume every open workshop form or Test Scene arrangement is included in a general session checkpoint.

05. Scripts and the script directory

Open Scripts in Simple or Advanced and select a file in Script files. A C# profile displays its `.cs` source read-only; save, validation, replacement and backup controls for existing source stay disabled. Extracting a C# source ZIP makes its files available to the server compiler or an external editor, not to Studio's Sphere editor. Use the target-specific generator for new C# classes and compile them with the actual server. The editing instructions below apply to Sphere `.scp` files.

Edit one definition

In a Sphere profile, select an item or creature and open Scripts. This shows the chosen section, not necessarily the whole source file. Make the edit, then use Validate and Save or Ctrl+S. The source path tells you which file will be updated.

Validation checks the section boundary, changed headers, accidental additional sections, supported control-block structure and whether the original has changed on disk. Warnings about legacy syntax need interpretation; Studio is not the Sphere interpreter. Keep unrelated sections out of a section-only edit.

Reload reads the current disk content. Resolve an unsaved-edit prompt before discarding work. Script editing preserves supported encoding and line-ending information. If a file has been changed by another editor, reload and reapply the intended change rather than overwriting the newer version.

Find text in the displayed script

Ctrl+F focuses Find. Type literal text; matching spans are highlighted and the count is shown. F3 moves forward, Shift+F3 backward. The arrow buttons do the same. Aa makes the search case-sensitive. Clear removes the search term. In the Find entry, Enter and Shift+Enter navigate; Escape returns focus to the editor.

This search covers the displayed text. To search other files, use the script workspace's Find / replace across files.

Open and create files

The right-hand Script directory follows the active profile. Double-click a .scp or select it and click Open file to edit the complete document. Refresh updates the tree. The New button or right-click menu offers New folder and New .scp file under the selected location.

Use clear filenames and categories. Sphere resource filenames must remain unique across the script tree for generated content. Creating a file in an arbitrary folder is not the same as adding it to Sphere's resource loading configuration; confirm that your server loads it. Studio-generated imports register their new script when needed.

Delete a script deliberately

Right-click a .scp and choose Delete script to review deletion of the whole file. Read its contents and the removal list before choosing Remove listed content. This operation keeps a backup. It retains other scripts and client assets; references elsewhere may need updating.

Remove empty .scp file is a separate action for a script with no remaining meaningful content. Use it after inspecting a leftover file. If your aim is to remove an item/creature and its assets, use Remove related content instead; chapter 24 explains its scopes and limits.

Work with multiple script tabs

Open All tools > Sphere script workspace. Open script selects a file inside the active server. Open current brings in the current script. Open more files to work in tabs; Close tab closes the selected document after handling its edits.

Ctrl+Space offers definition and Sphere-keyword completion. F12 or Go to definition follows a supported symbol; choose the correct target if there is more than one. Find references searches for the relevant symbol. Completion and reference navigation help with source editing but do not prove that code will run correctly.

Use Review save all to inspect changed workspace documents together. The main Save All also includes changed workspace tabs. A reviewed or queued save leaves this workspace open; check the queue if an older draft conflicts with your current text.

Replace text across selected files

1. Open Find / replace across files in the script workspace.
2. Enter Find literal text and Replace with. This is literal, case-sensitive replacement, not regular-expression matching.
3. Set Path filter, for example `scripts/**/*.scp`, or narrow it to the relevant folder.
4. Click Find changes. Select a result to inspect its before/after difference.
5. Select only the files you intend to change; Ctrl/Shift selects multiple rows.
6. Click Review selected replacements and read the complete proposed changes.
7. Queue or save the reviewed operation.

This tool reads saved script files. Save or reconcile any open drafts for those files first. A matching word can occur in comments or unrelated contexts, so inspect each selected file rather than treating an occurrence count as approval.

06. IDs, file formats and artwork preparation

Identify the kind of number

Number	Meaning
Item art ID	Ground/backpack artwork; its TileData record contains item metadata
Body ID	A creature, human or equipment animation body
Raw body and MUL bank	Physical animation destination within a particular bank
Sphere body ID	The body number used by the server, potentially mapped to another bank
Gump ID	Interface or paperdoll artwork
Hue ID	A colour table; game hue 0 means original colours
Multi ID	A compiled house/structure layout
DEFNAME	A named Sphere identifier, such as i_example or c_example

These are separate namespaces. Item 400 and body 400 are not interchangeable. Prefer explicit `0x` prefixes when entering hexadecimal. Fields labelled hex can interpret bare input differently from fields labelled decimal. `0x0190` and decimal `400` describe the same number, but are examples rather than suggested destinations.

MUL, UOP and Auto

MUL assets commonly use a data file plus a matching index, for example art.mul with artidx.mul. UOP stores entries inside an archive. Auto selects the format supported by that tool and found in the chosen client folder; it is not an instruction to update every format present.

Item artwork Auto normally prefers ArtLegacyMUL.uop when present. Animation Auto follows supported client mappings and available sources. Use an explicit source to investigate a mismatch. House compilation is a special case: it updates both MUL and UOP multi representations when both are present.

Editing a MUL while your client reads the corresponding UOP can leave the game looking unchanged. Check the review's affected files and the actual data location used by the recipient's client. Loose overrides, Verdata and client-specific mappings may also take precedence.

Prepare images and animation files

Use PNG/BMP for still art and supported image sequences; use GIF where the importer offers it. Preserve transparency and avoid painting a checkerboard into the image. Native UO artwork uses reduced colour precision and, in these import workflows, binary transparency. Inspect the converted preview: it is more relevant than the original full-colour source.

Keep animation frame canvases and ground origins consistent. A body VD contains supported action/direction data and palettes; a GIF or PNG sequence supplies frames, not all the behaviours of a complete creature. GIF timing does not become a MUL animation timing field.

Wearables need an inventory icon, equipped movement animation and paperdoll images. Paperdoll pieces must retain the complete intended canvas and alignment. Cropping the paperdoll tightly around the clothing can move it to the wrong place on the body.

Names and capacities

Client names often have byte limits, not unlimited text length: TileData and hue names use 20-byte fields; sound-import names have a 39-byte Windows-1252 limit. Use short supported names when a converter reports an encoding or length error.

Most formats use the allocation already present in your data. Equipment import has a specific opt-in expansion option, described in chapter 13. It does not make every archive or metadata table unlimited. A free candidate is evidence from the checked files, not a global promise that no world object or dynamic script uses that number.

07. Item artwork, TileData and the paperdoll

Inspect an item

Open Item artwork. Enter an Art ID (hex) and click Inspect, or select a Sphere item and click Use selected item. The latter follows supported item aliases, such as ID/DUPEITEM links. The view shows artwork and the client TileData record.

Export PNG saves the available item image. Copy art ID copies its number for another tool. Browse slots opens the allocation browser. Right-click the image for replacement, Show in Test Scene, Open asset workspace, Find references, Favourite / unfavourite, Export PNG or Detach preview.

Client metadata and Sphere properties are different layers. For example, a client tile's displayed name or flags do not establish every gameplay property of a scripted item. Check both Item artwork and Definition/Sphere scripts.

Edit the image in a graphics editor

Select an item image in Workspace and choose **Edit artwork**, or right-click the image and choose **Edit in graphics editor**. Choose an installed graphics editor if needed, then click **Open graphics editor**. Studio exports the original colours to a separate `artwork.png` copy. Save your changes to that same file; keep `asset.json` unchanged.

The returned preview appears automatically. Choose **Review edited image** to compare the original with the image after UO colour conversion, then use the normal Save or Queue controls. The same asset ID is retained. TileData, animation links, hues and scripts are retained. Edits do not install automatically. If the source files or project changed, reload the asset and start a new edit. A later editor save must appear in the preview before it can be reviewed.

Edit TileData

1. Inspect the correct item ID and choose Edit TileData.
2. In Item properties, adjust the fields you understand: name, weight, height, quality/equipment layer and animation ID.
3. Open Choose TileData flags to inspect named flags and descriptions.
4. Use the other stored-fields tab only when you need the specific raw value; preserve unknown fields.
5. Review changes, checking the ID and changed fields, then queue or save.

The flag picker supports name search, descriptions, double-click or Space to toggle, a full hexadecimal mask and additive presets. Presets include equipment, containers, surfaces, blocking decorations and animated statics. Additive means existing bits stay set: inspect the final mask and remove incompatible flags explicitly.

Weight 255 has special immovable conventions. Height affects geometry; equipment layer and animation affect wearing. Do not convert a decorative item into working equipment by changing only one field. Use the equipment importer when creating a complete wearable.

The small paperdoll

For supported equippable items, Item artwork shows Equipped item / paperdoll. Choose Male or Female and enter a hue; press Enter to refresh where required. This uses the linked paperdoll art, not the backpack icon.

Wear in Test Scene loads the selected wearable on the human model and shows the paperdoll alongside it. The paperdoll is a one-item inspection tool. Use Equipment / mount preview for manually combining multiple animation layers.

If the paperdoll is missing, inspect the item's equipment animation link and the corresponding male/female gumps. For the standard equipment link, male and female gump IDs are animation ID + 50000 and + 60000. These are linked destinations, not arbitrary empty gump slots. Queued supported TileData/gump changes can be included in the preview, so refresh after changing the queue.

08. Asset spaces and choosing destinations

Asset spaces is the gallery for item and animation allocations. Choose Items or Animations, then a source. Reload reads current saved files and pending reservations. Search by name or ID and filter All, Free candidates, Used, Referenced, Queued, Unavailable or the applicable No UOP entry state.

In a C# profile, Studio does not interpret arbitrary C# code to discover every graphic reference. Native client files, metadata/mappings and queued reservations still protect the destinations it can verify. A blank-looking or Free slot can be used only by C# source that this inventory cannot interpret; check those server IDs yourself before importing or replacing graphics. Source browsing is not a complete C# ownership analysis.

Cards use a written status as well as colour. Hover a card for its reason; double-click it or use Open details to inspect the destination. Import into slot opens the relevant importer. Choosing a used or referenced destination does not approve replacement. Queued and unavailable destinations cannot be selected for another import.

Status	Meaning
Free	No graphics, protected reference or queued reservation was detected in the supported inventory. Studio checks again during review and save.
Used	Existing native graphics occupy this destination.
Referenced	Metadata, client mappings or indexed scripts protect this destination, even when its picture is blank.
Queued	A pending import reserves this destination. The card shows saved client artwork; queued changes have not been saved.
Unavailable	The destination is read-only, unsupported, invalid or cannot be verified for this import. Read its reason.

Animation galleries and UOP ranges

Use Animations > All animations > Body animations for the complete supported populated client-body catalog without choosing a body range or a Sphere definition. Asset spaces remains the allocation inspector for imports. A catalog row identifies readable source data; browsing it does not choose a destination, reserve a slot or approve replacement.

MUL animation galleries describe their bank's body allocations. Modern UOP bodies use hashed entries and are browsed in ranges. UOP start body and Go jump to a range; scrolling can extend the loaded inventory when the search is empty and All is selected. Filters describe the loaded range. No UOP entry does not mean the app can write a modern UOP body there.

Slot browsers inside importers

Browse slots and similar buttons open a more focused candidate list. Search ID / name / reason, choose All slots, Free candidates or Used slots, and move between pages. Use selected ID copies the destination into the importer. Read the reason a slot is considered occupied.

The allocation checks consider saved native graphics, metadata, saved and staged indexed script symbols, applicable client mappings and pending imports. Valid bounded art.def static endpoints are protected. Relevant Verdata overrides or invalid override directories make affected graphic destinations unavailable; valid unrelated map-only patches do not block graphic imports.

World saves, map placements and dynamic script expressions are not scanned. A free candidate describes the supported inventory; it does not prove that every runtime use on the shard is absent. Reference scans and knowledge of the shard add evidence before deliberately repurposing old content.

Automatic placement

Automatic selection skips occupied, referenced, queued and unavailable destinations, using the same effective inventory as the manual browsers. Saved graphics in another present native source still protect their logical destination. A focused item range selector accepts a complete currently free range of 1-4096 slots; this does not change separate importer limits.

Native animation IDs in some banks share physical index records. The allocator checks the complete compatible body span and protected overlaps, including actions absent from the supplied VD. Secondary-bank imports also check their logical Sphere body identity. An empty selected action or five blank facings do not establish that the whole body is free.

Leave automatic placement enabled for new content. Find free space restores automatic choice after manual editing. Review the suggested range before saving:

- Separate item/gump image batches use a consecutive range in automatic mode.
- Animated statics need consecutive artwork IDs with their relative frame order preserved.
- Complete VDs need a compatible whole body block.
- Mounts also need riding-memory and, when importing a new icon, icon destinations.
- New equipment appearances need the animation and both paperdoll destinations together.

If there are free individual slots but no sufficiently long block, a consecutive batch still cannot fit. Reduce a separate-image batch or choose another supported source/bank. Do not split an animated-static sequence casually. Equipment capacity expansion only solves the specific animation-index shortage covered later.

09. Add item: reuse art or export an import kit

All tools > New Sphere item is useful for a basic static item. The Import tools can directly prepare client-file imports.

Reuse existing artwork

1. Choose Use existing artwork.
2. Enter the item name, a unique script identifier, artwork ID, category, subsection and weight.
3. Choose the script folder and a unique .scp filename.
4. If the art is newly imported and truly has no existing numeric Sphere base, use the corresponding numeric-base option; do not introduce a duplicate base.
5. Open Review item and inspect the preview and generated definition.
6. Choose Create script when correct. Studio writes and opens the new script.

This creates a basic item definition using the chosen graphic. It does not author creature animation or supply the movement/paperdoll parts of equipment. When the server already defines that art base, use inheritance rather than creating a competing numeric definition.

Prepare new PNG for UOFiddler

Choose that mode, select a single PNG, and enter a candidate art ID, TileData name, weight, height and flags such as Blocks movement or Surface. Review the converted image and script, then Export import kit to a separate output folder.

The kit contains the image, item metadata, script and instructions for the external import. Exporting it does not patch the client's art/TileData or install the new script. Complete the kit's import steps against a working copy before enabling its script on the target server.

Find a free candidate in this starter wizard has narrower coverage than every other Studio tool. Its MUL-based check needs matching art.mul, artidx.mul and tiledata.mul; UOP-containing setups may require an extracted MUL working copy. A kit does not reserve its candidate forever. Recheck it at import time.

For a fully in-app direct import, choose Import Tiles or Import Statics on Home or Import artwork, described next. Keep the original PNG and kit if you intend to share or revise the artwork later.

10. Import items, animated statics and effects

Import artwork without a server script

Keep **Artwork only** selected above the tabs. Choose the images, GIF or animation source, inspect the native preview and free destinations, then choose **Review artwork**. Item/effect imports prepare art, TileData and AnimData as applicable; body imports prepare their selected native animation sequences. Script names, stats, targets and destinations are not required or validated in this mode. No server script or `spheretables.scp` entry is created, and a raw body import does not add client body remapping. Existing scripts remain unchanged. You can generate a separate script for the installed IDs later.

To include new source now, choose **Artwork and server script** and configure the **Server script** tab before reviewing. Turning that choice off again excludes the script from the new review; it does not remove a script already saved by an earlier operation.

Choose Sphere, RunUO, ServUO or ModernUO

The importers have a Server script tab with a Script target choice. Sphere generates .scp files for Sphere X. RunUO generates C# .cs files for the official RunUO 2.x source. ServUO generates C# .cs files for official pub57. RunUO and ServUO use the same language, but their APIs differ; choose the actual server so weapon callbacks and rideable objects use the correct template.

This choice covers Monster, Mount, Item, Effect, Clothing and Weapon imports. Complete armor sets have a Script target for entire set control in their main window, including pieces loaded together from a folder. Each piece keeps its own appearance links and gameplay settings.

Sphere uses your active project and registers a new script, or an existing destination when required, in spheretables.scp. For a C# target, choose the C# server / export folder: the root containing Scripts, or an empty directory for export. Studio creates the selected category folder under Scripts at save time. It uses the UOAS_ prefix for generated class names, rejects existing files and duplicate class names, and shows the destination and complete source in Review. C# files do not register in spheretables.scp.

Review, queueing, file-access checks and recovery cover the C# file together with any imported assets. Compile the result with the chosen server, restart as required and use the displayed GM creation command. Studio does not compile or start your server. Every player's compatible client also needs the matching assets and metadata.

For artwork already installed, open All tools > Generate server script (Sphere / RunUO / ServUO / ModernUO). Select the kind, enter the existing graphic/body ID, choose properties and a target, then Review script and destination. This creates a script without importing artwork. It still uses the working profile and linked client directory for its save journal. Use decimal IDs or an explicit 0x prefix.

Choose a script file and folder

The Server script controls use the same destination choices for items, visual effects, monsters, mounts, clothing and weapons. They are also available in Generate server script and the Add item workflow for existing artwork. Choose your Script target, then set Save script:

- New script is the default. Choose Folder and keep the suggested filename in Simple, or edit New script filename in Advanced. Sphere creates a new `.scp`; existing filenames are reported before saving.
- Add to existing script is available for Sphere. Use Browse beside Existing script to select a `.scp` inside the active server's scripts folder. Studio adds the new generated definitions and required helpers while retaining the original definitions and comments.

- New folder stages a subfolder beneath the selected Folder. Enter its name and check the proposed destination. The directory is created only by the reviewed save, together with its script. Browsing, cancelling a chooser or preparing an import writes no folders.

Review the complete resulting `.scp`, its exact path and required resource registration before Save or Queue changes. New sections go before a single final `[EOF]`. Existing script encoding and line endings are retained. Duplicate definition/helper names, changed source files and unsafe destinations are reported; choose another identifier or review again. Several queued additions to the same script are combined in one reviewed save. Queue/session restoration and save recovery retain the operation.

For a whole armor set, Add to existing script uses One file for whole set so all its independently wearable definitions and helpers go to the chosen file together. Chapter 14 explains the armor layout choice.

RunUO, ServUO and ModernUO keep New script and separate `.cs` files. Folder, Browse and New folder remain available for the selected C# server/export destination; Add to existing script is a Sphere option. A UOFiddler import-kit export remains a separate artwork package rather than a reviewed direct script install.

Content packs reject queued appends to existing script files so a portable bundle does not copy unrelated old definitions. Choose New script when preparing generated Sphere scripts for Content packs; chapter 28 covers distribution.

Save a script recipe and preview the source

The standalone Generate server script window has Open recipe, Save recipe and Preview code above its tabs. Configure the asset, target, names, stats and rules, then choose Save recipe. Save the .uoscript.json outside your linked game folders. It keeps the graphic/body ID, mount IDs, equipment properties, bonuses and creature abilities that have been added to their lists. An unfinished rule still in its entry fields is not included; choose Add or Update first.

Open recipe restores those settings. Change the display name, identifier and artwork IDs when creating another asset. Recipe files contain data only, not executable scripts. Server/export paths, destination folders and filenames are excluded; choose the destination for the current project. Recipes are limited to 64 KB and unsupported or invalid settings are rejected before replacing your controls. These recipe buttons belong to the standalone generator, not the import wizards.

Preview code shows a read-only snapshot of the generated source and creation commands. Copy code copies that source. The preview does not save, compile or run it. Close the preview and choose Review script and destination when ready: the normal save review rechecks current files, duplicate names and the chosen destination. For Sphere, a loaded script project is needed to resolve existing definitions. For C# targets, native compilation and shard testing are still required.

ModernUO preview target

ModernUO also uses C#, but its constructors, combat callbacks and serialization differ from RunUO and ServUO. Select ModernUO in the same Script target control. The reference is official release 0.15.6.198.

Choose the ModernUO source checkout containing Projects/UOContent, or that UOContent folder itself. Studio creates category subfolders under Projects/UOContent/Scripts. An empty export folder also works: copy the generated Scripts folders into the recipient content project before compilation. Name checks cover the whole content project, including native Items and Mobiles, and exclude bin/obj output.

Generated ModernUO classes use Constructible, partial classes and native generated serialization. Creature cooldowns use anchored timestamps; summon references are saved and owned spell-target timers stop on deletion. Keep ModernUO-generated migration files with the server source and follow its migration procedure before changing a deployed class.

This is a preview target checked against official source. A complete ModernUO build and native gameplay/serialization tests were not run for this release because its required packages were not present in the existing build environment. Compile and test in a separate ModernUO project before deployment. Studio does not install or compile server dependencies. The existing Sphere-only tools described below retain their scope.

C# behaviour and compatibility

C# monsters use BaseCreature; animal mounts use BaseMount; rideable objects use a native ethereal-style inventory item without an animal to tame or feed. Stats, skills, resistances, tameability and follower slots are generated. Diet controls accepted FavoriteFood; None does not disable native pet-loyalty rules.

Normal selected spells use native mana, Magery, cast timing and targeting. During combat, the generated scheduler considers a spell at most every four seconds and prioritizes Cure and healing below the chosen threshold. Triggered abilities retain their chance, cooldown, health threshold, range and line-of-sight checks. During combat rolls follow the server's AI decision frequency.

Instant C# procs have explicit preset strength: power 500 gives 50 damage or healing before native mitigation. Poison level uses power / 250, capped at four; paralysis lasts at least one second using power / 100; stat debuffs use power / 100 for power / 10 seconds; Mana Drain removes power / 10 mana. Fixed and proc healing do not heal poisoned targets. These procs use no mana or reagents. Use normal spells for Protection, Bless, Reactive Armor and Magic Reflection. Existing Sphere functions cannot run on C# servers and are rejected.

Summons use an existing C# BaseCreature class such as Skeleton. Preset names are mapped automatically. Generated code tracks living limits, saves cooldowns and summon references, uses native summon expiry, and removes children when the parent dies or is deleted. Minions cannot be tamed and cost no follower slots; generated abilities cannot recursively summon. Native summoning removes carried loot. An empty export directory cannot check whether a custom class exists; the recipient's compiler must resolve it.

C# clothing uses BaseClothing; armor pieces use BaseArmor with generic plate material; weapons use the selected native weapon family. Bonuses use native attributes, resistance fields and up to five skill-bonus slots. Expansion rules determine their effect. Legacy weapon speed is converted to modern seconds as 150 / speed, with a 0.25-second minimum. Select Armor piece to use an armor rating.

C# effect imports create a GM-only preview item and a public Play(Mobile) method for custom scripts. They create visual effects, not a complete damaging spell. Animated artwork still depends on the client's animation metadata.

Tools that remain Sphere-specific

The Sphere definition index and editable source tools, sound-link editor, related-definition removal, house/deed scripts, world entrance scripts and content-pack script remapping remain Sphere workflows. C# profiles provide source browsing, not those Sphere definition/editing tools. C# generation does not convert arbitrary existing scripts or make C# classes appear in the Sphere editor. Share generated .cs files and matching client IDs separately instead of remapping them through Sphere content packs. The following import chapters describe Sphere output unless they explicitly mention C#.

The templates were compiled against official RunUO 2.x and ServUO pub57 sources, with constructor and serialization checks. Different forks may need changes. Test Scene previews artwork; it does not execute server AI, combat or equipment bonuses. Verify native spell scheduling, taming, mounting and balance on a working-copy shard. SERVER_SCRIPTS.md in the source package records detailed compatibility notes and exact reference revisions.

Choose the entry point

Choose Import Tiles on Home or Import artwork for floors, walls and roofs, or Import Statics for items, furniture and decorations. Both open the same item-art workflow. For moving objects choose Import Animation > Animated items; for spell artwork choose Import Animation > Spell / visual effects. These choices also appear on Import artwork and All tools. Both animation choices start with one animated static enabled and Artwork only selected. Choose Artwork and server script when you also need an item definition or visual-effect function. Item artwork > Import images / animated item opens the general art workflow; inspect its animation and script settings.

To inspect an already-saved native item/effect sequence, use Animations > All animations > Animated items & effects. Double-click it to play the stored frame order or export a GIF/PNG in the viewer. Chapter 15 explains those controls; the optional Test Scene preview adds a scene around the artwork.

Import one item or several independent items

1. Click Choose images for PNG/BMP files, Open folder for a tile folder, or Open ZIP for an image archive. Select a row to inspect its converted preview.
2. Choose Art files: Auto, MUL or UOP.
3. Leave automatic placement enabled, or deliberately choose the first art ID and mapping mode.
4. Use Move up / Move down to establish import order. Select unwanted rows and choose Remove selected to remove them from this import; the source files stay on disk. Read the destination list in the review.
5. Leave Make these frames one animated static item off for independent items.
6. Decide whether to Set TileData name, weight, height and flags for the listed IDs. Blank name uses the filename; other stored fields remain unchanged.
7. Keep Artwork only, or choose Artwork and server script and set the target, name, category and destination on Server script.
8. Click Review artwork, or Review files and script when source is included. Verify every proposed native ID and any definition before Queue changes or Save original files.

Consecutive IDs maps rows to a continuous range. IDs from filenames uses the file naming convention instead; use clear numeric filenames and inspect the proposed mapping. It switches away from automatic allocation. Do not assume a package's original reference IDs are vacant destinations on your shard.

Replacing existing artwork is an explicit choice. Enable Replace existing artwork at the listed IDs only when those are the assets you intend to change. For one replacement, the right-click Replace ... with image action is often clearer because it starts from an inspected ID and shows before/after art.

Import a static tile folder or ZIP

Choose Import Tiles or Import Statics on Home, Import artwork or All tools, then use Open folder or Open ZIP. The same source buttons are available in the item artwork importer. This imports static walls, floors and items; ground terrain uses a different format and belongs in World Generator. Choose Include subfolders before opening the source. PNG and BMP images are listed in natural filename order within their folders (tile2 before tile10). Other file types are skipped; ZIPs are read without extraction.

Keep Include subfolders enabled for a downloaded pack with several artwork folders. By default, Studio excludes subfolders named previews, regardless of capitalisation, and filenames marked with separate preview, previews or contact-sheet tokens. Oversized overview sheets and other incompatible images are reported and skipped while compatible single-frame PNG/BMP tiles load. Studio does not resize these images or split a sheet into tiles.

PNG is preferred over a matching BMP in the same folder. This also applies to exact mirrored format folders: for example, `png/walls/tile1.png` replaces the alternative `bmp/walls/tile1.bmp`. The pair must have exactly one matching-format directory component at the same depth; the remaining folders and filename stem must match. Unrelated paths remain separate. If the preferred PNG is invalid, Studio reports it instead of silently substituting the BMP.

The source status shows the loaded count and skipped-file counts. Choose View skipped files... to read or copy filenames and reasons in a scrolling report. If a pack has no compatible tiles, the previous selection remains and Studio shows the new pack's report. Review the loaded list before choosing destinations; a successful load can omit incompatible images.

Choose Advanced to show Include preview images and Include both PNG and BMP versions. The preview choice bypasses the folder/filename role filters; preview images must still satisfy dimensions and native-format limits. The variant choice keeps both PNG and BMP alternatives. Set these choices before reopening the folder or ZIP; changing a checkbox does not replace the loaded list. Choose images provides an explicit file selection with its existing validation, rather than the pack filters. Remove selected removes unwanted rows from the import list without deleting source files.

Static folder and ZIP batches support up to 4096 compatible still tiles, each 1-512 pixels per dimension, within 16 million total pixels and 128 MB of input. Each image file must be within the 16 MB limit. Direct Choose images retains a separate 256-file cap; the larger pack limit does not expand native animation sequences. Resource limits, cancellation, unsafe or duplicate ZIP entries and password protection stop the complete load without replacing the previous selection. Split packs that exceed a batch budget. Corrupt or native-incompatible static images are listed in the skipped-file report.

Folder names remain visible in the list. Blank TileData names use each image's filename without its folder or extension. IDs from filenames reads the final filename: for example, `walls/0x4000.png` requests item art ID 0x4000. Duplicate destination IDs are rejected even if the files are in different folders. Automatic placement instead finds a consecutive free range, taking queued changes into account.

For independent tiles, leave the animated-static option off. Set the appropriate TileData collision, surface, height and other flags; the importer cannot infer a wall's intended behaviour from its pixels. The chosen settings apply to the batch, so import walls and floors separately when their properties differ. Review every ID and proposed file change before saving. Loading a folder or ZIP alone changes no client files.

Being under the 512-pixel dimension limit alone does not guarantee that a densely filled image fits the native static-art format. Folder/ZIP loading checks that compatibility before keeping a tile. If Review identifies an incompatible explicitly selected image, read its filename, remove the reference sheet or prepare compatible artwork, then review again. TileData collision, ladders and other gameplay behaviour are not inferred from a pack's preview or manifest.

Make an animated static

An animated static is item art driven by animdata.mul, not a creature body animation. Examples include a decorative animated object or frames for a visual effect.

Choose Import Animation > Animated items or Spell / visual effects, then load a GIF or ordered PNG/BMP sequence. Make these frames one animated static item starts enabled and stays enabled when loading a frame folder or ZIP; preview folders and duplicate BMP alternatives are skipped by the normal source filters. In the general item-art importer, enable this option yourself. Use 1-64 consecutive artwork IDs in frame order. Play frames checks order at the tool's preview rate. Set Frame interval to 1-255 client ticks. FrameStart accepts 0-255; it is an advanced stored field, not a general GIF start-time selector. Leave its default unless you are reproducing a known arrangement.

Animated item/effect folder and ZIP loads stop on an incompatible selected frame rather than silently removing it and changing the sequence. The static filename-role filter does not remove animation frames named with preview or contact-sheet tokens; the preview-folder and PNG/BMP choices still apply. Correct the named frame or prepare a complete compatible sequence, then reopen the source. If a static load already excluded preview or incompatible images, reopen it with animation enabled before making it an animated item; Studio blocks reusing that filtered list as a sequence. The native animated-static limit remains 64 frames; a pack's 4096-tile limit is for independent static art.

The reviewed save includes the relevant art, TileData and AnimData changes. An animated item gets its main script definition at the first artwork ID; independent images get separate definitions. The native AnimData sequence has a supported frame-count/offset limit, so a long GIF may need deliberate preparation rather than direct import of every frame. Studio reports unsupported sequences before saving.

Generate an effect script

Choose Import Animation > Spell / visual effects, or the same choice in All tools. The art setup follows the animated-static workflow and starts with Artwork only. Choose Artwork and server script when a callable visual function is wanted. On Server script, choose Follow character or Ground / object and set Effect speed and Loop duration. These are fields for the generated effect call; they are distinct from frame playback and image timing.

Studio generates `f_<identifier>` and the necessary item definition, using the first artwork ID. The review shows the actual script and test command; use that command on a suitable test shard after saving/loading the files. Test Scene can preview ground, on-player and projectile visuals, but it does not run the generated function.

The effect importer creates a visual effect. It does not automatically design damage, targeting rules, cooldowns, team filters or a complete spell. Add those server behaviours deliberately in Sphere scripts. If the art already has a numeric base, inspect the generated ID inheritance so the new named definition uses that base rather than duplicating it.

Generated script fields

Simple derives the script identifier and suggested new filename from the display name and chooses the usual folder. Save script, Folder, Browse and New folder remain available in both modes. Choose Advanced to customise the identifier, category, subsection or New script filename. For Sphere, Add to existing script lets you choose an existing .scp. Review the complete resulting source and destination before saving.

Display name is the player-facing text. Identifier supplies the named script symbol; use letters, digits and underscores with a sensible prefix supplied by the importer. Category and Subsection organise the definition list. Script folder and New .scp filename identify the output. Scripts default to type-specific folders and are registered as resources when needed.

For monsters/mounts the same panel includes basic STR/hits, DEX and INT fields. For equipment, gameplay fields come from Item and icon. Review the full generated script rather than assuming a template fits every shard's custom systems.

11. Import monsters and other characters

Open Creature or monster on Import artwork or All tools. The animation layout describes how frames are stored; the generated CHARDEF supplies gameplay properties.

For body-data import without an initial creature script, choose Import Animation > Body / character animations on Home or Import artwork and keep Artwork only selected. It accepts VD, GIF or image frames. A VD supplies its stored action/direction sequences; a GIF or frame list supplies one selected action and facing. Neither source creates missing directions. This also imports raw mount body data without configuring a rideable creature. Choose Artwork and server script when you also need a definition and client body mapping, or use the complete Mount workflow for its inventory/riding setup.

Choose a source and destination

Choose frames, GIF or .vd, or choose Open ZIP for a downloaded animation package. Studio offers the packaged native VDs when present; choose the intended one if several are listed. Otherwise choose a GIF or one same-folder PNG/BMP sequence. Studio does not combine unrelated animations. PNG/BMP sequences and GIFs supply one action and stored direction. A compatible VD supplies its populated sequences and layout. Use the animation workshop for frames.json or body.json exports so their registration metadata is retained. Review the sequence list before choosing a destination.

Save loaded animation as .vd exports the supplied animation without importing it into client files. A loaded compatible VD retains its populated native sequences; loaded images/GIF produce the single sequence you configured. The completion message reports what was included. It does not add missing actions or directions. Chapter 16 explains the conversion and transparency limits.

Field	What to check
MUL bank	The physical animation files being written, banks 1 through 6
Raw body ID	The destination inside that bank
Action	The animation action for a frame-sequence import
Stored direction	One of 0 through 4; the remaining display directions use mirroring
Animation file layout	13 Animal, 22 Monster or 35 Human actions; VD determines its own format
Sphere body ID	The server-facing number; secondary banks may require linked mappings
Origin X / Origin Y	Frame registration for image imports; use the preview to align the ground

Keep automatic free-block selection on for new content. Browse body slots allows deliberate selection. Bank 1 normally uses the same raw and Sphere body ID; secondary-bank imports can prepare the necessary mappings with the generated script workflow. Confirm those changes in the review.

Import a complete body

The destination must have a compatible whole native body allocation. Existing graphics, metadata, mapped identities, overlapping body spans and pending imports protect it. Use automatic placement for a new body; intentional supported replacement needs the explicit confirmation described in chapter 4.

1. Load the compatible VD.

2. Check its layout and populated action/direction sequences.
3. Choose a suitable bank and Find free space.
4. Preview several directions and actions, especially stand, walk, run and required combat/death actions.
5. Choose Artwork and server script and set a unique name, identifier and destination script. Keep Artwork only when importing the body data alone.
6. Configure Creature behaviour, then review all native files, mappings, the CHARDEF and any ability helper functions.
7. Queue/save, then test the character on the target shard.

Import a single sequence

Choose the image sequence or GIF, assign the intended action and stored direction, and set a compatible body layout. Adjust origins, update the preview and reorder frames where necessary. Import other required directions/actions separately.

A walking GIF in one direction is not a complete monster. The template can create a basic walking monster definition with combat properties and a monster brain, but missing movement/attack/death art remains missing. Preview rate does not alter the server/client movement timing.

Creature behaviour: spells, taming and special attacks

Choose Artwork and server script in a Monster import to enable Creature behaviour and include these choices in the new script. The complete Mount workflow already includes its required script. This configures a new imported creature; it does not edit existing creature definitions. STR / hits, DEX and INT / mana remain on Server script.

Apply preset offers Basic melee, Tameable animal, Armored fighter, Spellcaster, Self-healing fighter, Poisonous creature, Dragon-style pet and Lich-style summoner. Applying a preset replaces the current behaviour settings and ability list; caster presets also set INT to 100. Every value remains editable.

Page	Controls and purpose
Stats, taming & food	Animal or Monster brain; minimum/maximum damage, armor, fame and karma; Wrestling, Tactics, Magic resistance, Magery, Evaluating intelligence and Meditation; physical, fire, cold, poison and energy resistances
Taming and feeding	Can be tamed, required taming skill, follower slots and Meat, Plants, Omnivore or None diet. None disables hunger. Taming a monster does not add riding animations; use Mount import for a rideable creature
Spells & healing	Casts spells using normal Sphere AI; choose offensive, healing, cure and defensive spells. Set Heal at or below HP %. Magery must be above zero; mana, spell strength and normal casting are governed by the shard
Triggered abilities	Add ability, select/edit, Update selected, Remove selected and New / deselect. Up to 12 rules, each with its own trigger, chance, cooldown, own-health threshold and enemy range

During combat rolls when Sphere makes a combat decision, not at a fixed rate per second. On a landed hit fires after a successful melee hit check. When hurt fires during incoming damage handling, before that damage is subtracted from HP. Health thresholds always refer to the creature's own health. For example, HP at / below 40% makes a rule eligible only while it is at or below 40% health. Enemy abilities check distance and line of sight.

Ability	What it generates
Spell effect	An instant spell proc, such as Flamestrike, Fireball, Poison or Paralyze against the enemy, or a beneficial spell on self. Power 500 means 50.0 skill units. It uses no mana, reagents or casting delay
Heal self	Restores a fixed amount of the creature's HP, capped at maximum. Set a low-health threshold and cooldown. This uses no mana or bandages
Visual / damage effect	An existing client effect art ID, shown On target, as a Projectile or on the Ground. Optional sound ID and physical damage. Zero damage is cosmetic; self visuals require zero damage
Summon creatures	Existing CHARDEF such as c_skeleton, number per activation, maximum living summons and lifetime. Temporary allies belong to the summoner, cannot be tamed and expire automatically
Existing function	A creature-compatible Sphere function selected from the linked server, such as a suitable f_fire_nova. It runs on the creature with the enemy as SRC and ARGN1; that function controls targeting, visuals and damage

To make a lich-style summoner, apply Lich-style summoner, open Triggered abilities, select the summon rule and adjust the creature, chance, cooldown, count, limit and lifetime. Choose Update selected to keep the edit. Stock lich behaviour differs between shard script sets; this preset is a starting point you control.

Each summon rule creates 1-5 creatures per activation, allows up to eight living children and gives them a 5-600 second lifetime. The combined living limits across rules cannot exceed 12 per summoner. Summons are removed when the summoner dies or is deleted, and do not consume follower slots. They retain the chosen existing creature's shard scripts, which may add other behaviours. Studio-generated triggered abilities are suppressed on generated minions to prevent recursive summoning.

Review files and script shows the complete CHARDEF and helper functions before saving. They stay together in the monster or mount script and through queued saves/checkpoints. Effect art IDs and sound IDs must already exist in the client; this page does not import separate effect artwork or audio. A listed function must be suitable for NPC use. Native spell selection, combat balance, resistance rules and mounted actions need testing on a working-copy shard: the in-app Test Scene previews artwork and movement without running Sphere combat AI.

Replace part of an existing body

The importer checkbox prepares a replacement proposal. It does not grant overwrite consent. Inspect **View graphics review details…** in the final review and confirm the exact affected graphics. Native span aliases and linked logical body identities are still checked; queued or unsupported destinations stay blocked.

Inspect the existing body and source first. Enable Replace the listed destination sequences only for an intentional update. Unlisted actions/directions remain unchanged; this is useful for targeted corrections but can also leave a body with mixed old/new artwork. Check the sequence list and the final body in multiple actions.

Original VD transfer preserves its supported palettes and origins. PNG conversion can requantise colour. Modern UOP body animation is browsed/exported, while this writer targets supported MUL sequences; it does not provide automatic full modern-UOP-to-MUL conversion.

12. Mounts, rideable objects and inventory icons

Choose Import Mounts on Home, Import artwork or All tools. It opens the mount workflow directly. A working mount combines body animation, server definition, riding-memory item/mapping and inventory/shrink artwork. Each part has its own role; an icon does not supply a mounted pose.

Import a mount

A mount links several destinations. The body, riding-memory item and any newly imported icon are checked together against saved graphics, references and pending reservations. A manual choice remains subject to the same final replacement review as an automatic choice.

1. Choose mount .vd, or use Open ZIP and select the intended packaged VD. Inspect the supplied layout.
2. Select the MUL bank and leave Automatically use next free animation block and linked IDs enabled.
3. Find free space. Review raw/Sphere body, mount-memory and new icon destinations.
4. Set the memory item's short client name. Do not reuse a referenced memory record without deliberately enabling and reviewing replacement.
5. Choose the inventory image on Shrink icon.
6. On Server script choose Animal or Rideable object and complete the script fields.
7. Preview actions/directions and open Review files and script.
8. Save and use the generated test command to create the correct object on your test shard.

Compatible 35-action mounts retain the human-style layout and require the relevant unmounted walk/run/stand and mounted walk/run/stand sequences. For that layout, the required pose groups include actions 0/2/4 and 23/24/25 in five stored directions. Other supported layouts have their own checks. A layout named Human does not force the mount to behave like a human character.

Animal or Rideable object

Animal is the default template for a tameable mount. Rideable object is for a unicycle or contraption without taming/feeding. It still uses Sphere's character/mount machinery internally, with ownership and no autonomous wandering/following in the generated object workflow.

For a rideable object, use the generated `.add i_ride_...` command shown in the review. Put one item directly in the character's backpack and double-click it to mount immediately. Dismounting returns the same item to the backpack. Sphere hides the original item while ridden and removes its temporary backing creature on dismount; repeated clicks cannot create another mount. The item is blessed and keeps its hue. RunUO, ServUO and ModernUO use their native EtherealMount class. Test the complete cycle on your shard, especially custom death, trading and mounting hooks. Simply spawning the creature body bypasses the intended item workflow.

This choice generates the new script. It does not automatically convert earlier installed animal scripts or change an already deployed world object.

Choose a shrink or inventory icon

Shrink icon offers existing artwork, a new PNG/BMP, an explicitly chosen image from a ZIP through Choose image from ZIP, or a frame from the loaded mount animation. Existing artwork can be searched by name, DEFNAME, category or ID, with Figurines first / Figurines only / All artwork filters. Find horses helps locate existing horse-like figurines.

For a new image, choose a separate free art slot and a short client name. The riding-memory item and the new visible icon must not be the same destination. Reusing shared icon art does not mean you should overwrite its existing Sphere definition.

Turn a pose into the icon

Select Use animation frame, then Choose pose. Choose an action/direction and a frame thumbnail. Set Fit within pixels, Trim transparent edges and Mirror pose if required. Inspect the converted still image, then Use this frame as icon.

The selected frame becomes still item artwork saved with the mount; no intermediate PNG export is necessary. Changing the source VD clears the previous frame choice to avoid accidentally keeping an icon from another mount.

Check the completed mount

Test standing and travelling in every direction, the rider's seat, overlap with clothes and the inventory/shrink image. Studio's rider offsets are preview controls; changing them is not a saved animation correction. Use the animation tools or source artwork to make a persistent registration change. Ownership and feeding rules need server testing; Test Scene checks appearance only.

13. Clothing and weapons

Open All tools > Clothing or weapon. The five tabs are Item and icon, Equipped appearance, Animation preview, Bonuses and effects, and Server script. Use this workflow for one independently equippable piece.

Understand the required assets

Part	Purpose
Inventory icon	The item on the ground or in a backpack
Equipment animation	How the worn item looks on the moving world character
Male paperdoll image	The item overlaid on the male paperdoll
Female paperdoll image	The item overlaid on the female paperdoll
TileData and script	Links those assets to the layer and gameplay properties

One image cannot stand in for all of them. A whole-suit animation is one appearance; it cannot automatically be divided into a helmet, chest and boots that can be worn separately.

Item and icon

Choose existing equipment to copy when you want a starting point. Check the new destination item ID, short client name and Artwork files source. The copied appearance/icon is a source; the destination should still be deliberate.

Choose the Equipment layer and set weight, required strength and durability. Clothing has Armor, Armor piece (Sphere t_armor) and Dyeable in Sphere. Weapons have Weapon type, minimum/maximum damage and Sphere weapon speed. Partial hue limits colouring to grey pixels where supported; it is separate from the preview hue selector.

Icon source Import reads a PNG/BMP. Copy reads another item ID. Preview icon shows the backpack/ground artwork after conversion. Check that a helmet's icon is not accidentally being reused for every piece of a set.

Reuse an equipped appearance

On Equipped appearance, choose Reuse and select the existing equipment animation ID. Studio reads its animation and linked male/female gumps. This is useful when making a new scripted item with an established appearance.

Choose the paperdoll source matching the recipient's files. Load appearance and check action coverage, then inspect the male/female images. Reuse does not mean the existing shared animation is copied to a new body or rewritten.

Import a new equipped appearance

Studio checks the linked icon, equipment body and male/female paperdoll destinations together. Existing native graphics, aliases, references and queued imports remain protected. If a supported intentional replacement affects existing graphics, confirm that exact set in the final review and retain its default backup.

Choose Import. Supply a compatible 35-action human/equipment VD and both paperdoll PNG/BMP files. Preserve the full canvas and transparency of the paperdoll pieces. The equipment animation links to male gump `50000 + animation ID` and female gump `60000 + animation ID`.

Keep Next free animation and paperdoll bundle enabled, or use Free animation + gumps for an intentional destination. Both linked paperdoll slots must be available as well as the animation body. Load piece folder fills the icon, VD and paperdolls together from a correctly named folder; Open piece ZIP finds those four conventional files together inside the archive. Select the intended piece if several are found, then check its name and layer yourself. To import a whole set, use Complete armor set. Chapter 14 gives the file convention.

Action coverage and gameplay fields

Open Animation preview and Load / refresh appearance. Select an action, direction and preview rate, then Show sequence or Play. The table shows which of the five stored directions are present. The shared cross marks the registration origin; the equipment layer is shown alone here.

Allow incomplete action set is an advanced option after you have reviewed missing sequences. It does not waive required walk/run/stand coverage or a weapon's required attack. For a release-quality wearable, inspect movement, riding, attack and both character/paperdoll variants as applicable.

Optional bonuses, penalties and on-hit spells

Before Review complete import, open Bonuses and effects. Select a preset, enter its Amount or Chance %, then choose Add / update effect. Selecting an existing row loads its values for editing. Adding the same preset updates it instead of stacking duplicate rows. Remove selected deletes one choice; Clear effects returns to ordinary equipment. The script preview shows the additions, and the final review shows them in the complete item definition.

Preset	Example and behaviour
Strength, Dexterity or Intelligence	-5 Strength lowers the wearer's strength while equipped; +5 increases it
Luck	+100 Luck while equipped; the shard's loot rules determine its benefit
Physical, Fire, Cold, Poison or Energy resistance	+10 increases that resistance while equipped; damage rules and caps belong to the server
Damage increase, Hit chance or Defence chance	Add an equipment combat modifier; the shard still controls the formula and caps
Skill bonus	Enter 5 for +5.0 effective skill or 2.5 for +2.5; up to five skills per item
On hit: Flamestrike, Fireball, Lightning, Harm, Magic Arrow, Energy Bolt or Explosion	Weapons only: choose 1-100% and spell power 1-1000

For example, choose On hit: Flamestrike, enter 15 in Chance %, and leave Spell power at 500. This adds a separate 15% roll to each successful weapon hit. Power is Sphere spell strength in tenths: 500 means 50.0. It is not guaranteed damage; the selected spell and shard rules determine the outcome. Each selected spell rolls independently, so more than one can activate on the same hit. The ordinary weapon damage continues.

These presets target Sphere X. Stats, Luck and resistances use its native equipment properties, which the engine adds and removes as equipment changes. Elemental resistance requires the server's elemental combat rules; Studio does not change server configuration. Skill bonuses use temporary effective-skill modifiers with

matching unequip and item-removal cleanup. They preserve trained skill values and stack with other equipped preset items. Avoid editing a generated skill helper while its item is worn; unequip it before updating scripts.

The choices belong to the individual piece. In Armor set, Edit selected piece opens its saved effects. Copy selected piece's effects to all pieces replaces every piece's effect list with the selected list. This applies a bonus per piece, not once for wearing the complete set: copying +5 to eight pieces can total +40. Review that total before saving. Folder-imported pieces start with no optional effects.

Generated scripts remain editable in the normal Sphere scripts workspace after the reviewed import. The Test Scene and paperdoll display artwork; they do not execute these combat effects. Test equip, unequip, repeated swaps, skill stacking and spell balance on a separate server before distributing equipment.

On Server script, inspect the identifier, category and destination. Review complete import combines the new icon, TileData, imported appearance assets where applicable and script. It does not create missing artwork or automatically generate female animation substitutions.

Expand equipment capacity if needed

If no suitable primary equipment block exists, enable Expand equipment capacity if needed before finding space. Studio can append blank primary animation-index records for otherwise unused equipment IDs up to 2047. The expansion is shown in the review and occurs only when saving.

Existing references, mappings and occupied records stay protected. This option does not expand TileData, every animation bank or paperdoll allocations, and it does not author missing frames. You still need a free icon and both paperdoll destinations. Recovery of the reviewed operation restores the original index length when applicable.

14. Complete armor sets and whole-folder import

Open All tools > Complete armor set. A set groups independently wearable pieces into one reviewed save. Each piece still needs its own icon, animation appearance and paperdolls, or an explicitly reused compatible appearance.

Build a set manually

1. Enter Set name.
2. Choose a piece such as Helmet, Gorget, Chest, Arms, Gloves, Legs, Boots, Cloak or Robe.
3. Click Add piece. The equipment editor opens with an appropriate layer and armor category.
4. Fill in its icon, appearance and properties. Check animation coverage.
5. Choose Check and add piece to set to return to the set list.
6. Repeat for the remaining pieces.
7. Select any row to Edit selected piece or Remove selected piece.
8. Choose Review complete set and inspect all destinations and scripts before saving.

Adding or removing a row only changes the pending set form. It does not write or delete game files. Suggestions exclude pieces already in the set. Duplicate destinations, layers, animation IDs, identifiers and filenames are checked before accepting the combined import. Piece scripts normally use `scripts/armor`, Category Armor and the set name as Subsection.

Choose the Sphere script files

Script target for entire set selects one server target for the whole import. For Sphere, choose how to organise its definitions:

- One file per piece keeps the existing separate `.scp` files and each piece's reviewed destination. This is the default.
- One file for whole set puts all the set's item definitions and their required generated helpers into one `.scp`. Keep New script and choose its Folder and filename, or choose Add to existing script and Browse to the intended Sphere `.scp`. Adding a whole set to an existing file selects One file for whole set automatically.

For example, eight pieces can use eight individual scripts, one new `armor_example_set.scp`, or one existing armor `.scp` with the eight definitions added. The pieces remain independently wearable, with their own identifiers, layers and native appearance assets. This setting changes the script layout and destination, not the equipment artwork or per-piece effects.

New folder stages a subfolder for a new script; it is created only when saving the reviewed operation. Review complete set shows the selected layout, exact destination paths and complete resulting script text before Save or Queue changes. Existing-file additions retain the original definitions and comments and keep one final `[EOF]`. Sphere resource registration is included when needed. RunUO, ServUO and ModernUO retain their separate `.cs` files per piece.

Load a whole set at once

1. In All tools > Complete armor set, choose Open ZIP and select the downloaded armor set or collection ZIP. You do not need to extract it first.
2. Check the detected set name and piece count. If the archive contains several sets, select the intended set from the list.

3. Choose Add whole set. Studio loads the matching icon, equipment VD and male/female paperdolls for every listed piece, then proposes fresh destinations.
4. Edit the resulting rows when needed, then choose Review complete set and inspect every destination and generated script before saving.

For files you already extracted, choose Open set / collection folder and select the outer set folder, such as `armor-collection-example-set`, containing the piece folders and manifest.json. Parent collections and extraction wrappers can reveal their contained sets. Keep the downloaded layout together; you do not need to open each piece separately.

Existing manifest.json collections with a sets list remain supported. Website downloads may instead describe one set with pieces (folder and layer) or items (files and layer). Each piece still needs its four separate source assets. Source/reference IDs in the download remain references; Studio allocates destinations for your selected client.

Opening a parent collection reveals its contained sets; Add whole set adds the selected set rather than every collection entry. If a layer is already present in the current form, remove or edit that piece before adding another set with the same layer. Missing or ambiguous source assets are reported before adding the set; opening a folder does not write game files.

Enable Expand equipment capacity if needed in the set builder before loading pieces when the existing primary equipment allocation is too small. The option is carried into the piece requests and remains subject to the checks described in chapter 13.

Simple folder convention

A manifest is optional when the folder contains recognised named piece subfolders. Each piece folder needs exactly one of each of these assets; PNG image names may use BMP equivalents:

```
ExampleArmor/
  head/
    equipment.vd
    inventory-icon.png
    paperdoll-male.png
    paperdoll-female.png
  torso/
    equipment.vd
    inventory-icon.png
    paperdoll-male.png
    paperdoll-female.png
  gloves/
    equipment.vd
    inventory-icon.png
    paperdoll-male.png
    paperdoll-female.png
```

Recognised folder names include head/helmet/hood, gorget/neck, torso/chest, arms, gloves, legs/pants, boots/shoes, cloak, robe, waistcloth/belt, skirt and shirt. An unknown piece folder requires a manifest with its suggested layer or manual addition.

Manifest structure for a collection

For a custom collection manifest, use a sets list; each set has a name and pieces. Each piece names its suggested_layer and four relative file paths. For example, this describes one helmet using layer 6:

```
{
  "sets": [
    {
      "name": "Example Armor",
      "pieces": [
        {
```

```
          "name": "Helmet",
          "suggested_layer": 6,
          "files": {
            "inventory": "head/inventory-icon.png",
            "animation": "head/equipment.vd",
            "male_paperdoll": "head/paperdoll-male.png",
            "female_paperdoll": "head/paperdoll-female.png"
          }
        }
      ]
    }
  ]
}
```

Website per-set manifests with pieces/folder/layer or items/files/layer are recognised directly; you do not need to rewrite a downloaded manifest into the collection example above. Paths must stay inside the selected folder. Each manifest set supports 1 to 16 pieces and valid equipment layers. Do not put absolute personal paths in a distributed manifest. Original source/reference IDs are not automatically used as installation destinations.

Before distributing a set

Check each individual piece, then the intended combination: backpack icons, male/female paperdolls, world standing/walking/running, mounted poses and attacks where applicable. Confirm that the files represent separate pieces, not the same complete-suit VD copied repeatedly. Keep the source files and manifest so recipients can reallocate for their client.

15. The Animations viewer and exports

Open Animations from the sidebar or Ctrl+K. The Animation library starts in All animations and reads the active project's linked client files. You can browse saved animation data without entering a body ID or finding a Sphere definition first. Choose Body animations for creatures, mounts and equipped appearances, or Animated items & effects for the separate native artwork sequences.

Browsing and playback are read-only. They do not import data, reserve an ID, change client files or enable replacement. Import Animation still opens the appropriate importer with its usual preview, destination protection and reviewed save.

Browse body animations from the client

1. Confirm the active project's extracted client folder using Manage > Edit paths. Body sources need their matching anim*.mul/idx pairs or AnimationFrame*.uop archives.
2. Open Animations and keep All animations selected. Choose Body animations. Wait for the library's progress message to finish; UOP discovery can take longer on the first read.
3. Search by decimal ID, 0x hexadecimal ID, name or source text. For example, 71, 0x47 and 0x0047 identify the same raw ID where it exists. Names are optional client/shard metadata; unnamed bodies remain available.
4. Choose All sources, a MUL bank or UOP. The same raw body ID can contain different artwork in different banks. Thumbnails loads visible pictures as you scroll; List shows the IDs, sources and names as rows.
5. Double-click a card or list row, or select it and press Enter. Enter in the search field opens its first current match. The viewer chooses an available action and facing from that exact source.
6. Inspect Action, Direction and Zoom, then use the playback controls. Available actions depend on the selected facing. Directions 5-7 mirror their corresponding stored MUL facing when one exists; a missing facing is reported rather than generated.

Hover over the selection to read its exact source filename, raw body ID, optional logical client mappings, names and overlapping physical spans. Search can find those linked logical IDs as well. A raw bank/body identifies stored records; a logical client ID can route to another bank/body. Shard definitions limits the list to entries with resolved shard names. If that filter is empty, choose All animations to return to the client catalog.

Browse animated item and effect artwork

1. In the same library, choose Animated items & effects. Keep All animations to see supported nonempty AnimData sequences regardless of Sphere definitions.
2. Search the base item-art ID or name. All sources, MUL and UOP filter the effective artwork source. The automatic artwork preview uses ArtLegacyMUL.uop when present; otherwise it uses the complete art.mul/artidx.mul pair.
3. Hover over a sequence to inspect its stored frame IDs, native interval/start bytes and missing-artwork notices. Frame offsets can point before or after the base ID and can repeat a frame. The thumbnail uses the first actual frame in that order.
4. Double-click the result or press Enter. The Animation viewer loads its native AnimData frame order in this tab. Use Play/Pause, Previous, Next or the frame slider, then Save GIF or Export frame. Preview FPS, Speed, Zoom and Loop control this preview and the GIF settings.
5. To see the sequence against a floor or other assets, choose Preview in Test Scene. It opens a separate Effect preview with Effect frames set to 0, using AnimData.

AnimData does not identify whether a sequence is an ordinary animated item or a spell effect. The library does not infer combat behaviour from its ID or name. Stored interval and initial-delay bytes remain visible metadata; Studio uses your selected preview timing rather than promising the game's automatic-item or spell timing. Some sequences require explicit effect playback even when TileData's animated flag is off. Item artwork opens individual pictures; Animations plays their sequence. Chapter 10 covers importing item/effect frames and chapter 28 covers distributing their artwork, metadata and optional scripts together.

Save GIF and Export frame apply to body sequences and animated item/effect sequences. VD exports apply to body animations. Choose a Body animations result to use the body-specific action, direction and VD choices again.

Physical sources and client-routed bodies

Library body selections preview physical records from the chosen source. A MUL selection keeps its raw bank/body. A UOP selection shows physical archive group IDs directly, without AnimationSequence action aliases. Source text and the selected Action identify that raw preview; it can differ from the creature pose chosen by the game.

The manual Body ID field and Selected creature remain available. Enter a decimal or 0x hexadecimal logical body ID, choose Source: Auto, then Load body to follow the supported client routing. Selected creature fills the ID from the selected CHARDEF. The status identifies the resolved source/body. An unavailable action can mean that this routed body or facing does not contain it.

For a raw UOP library selection, Save entire body as .vd is disabled. Convert routed body to .vd and Routed body export options use the client's action layout, which can select different physical groups from the raw preview. The separate right-click Show routed body in Test Scene command also follows that routed interpretation. Export frame and Save GIF still save the exact sequence displayed in the raw viewer. Read the route and coverage reports before treating a routed VD as a complete body.

Reload and source notices

Reload library rereads the linked animation sources. Use Edit paths again when you have added or changed the set of client files linked to the project. A project switch discards the old results and thumbnails. A source change while reading rejects that result; choose Reload library to refresh the previews after the correct files are linked. Already displayed cached thumbnails can remain until you reload. No animation files linked means the current folder lacks the required extracted sources, not that you must create a Sphere definition.

The library reports incomplete MUL pairs, invalid indexed records, missing item frames and unsupported entries. Some optional naming/routing metadata can fail while physical body rows remain readable; routed conversion still requires its own valid metadata. Hover over source notices for the details.

Legacy art.def/verdata overrides are not applied by these native animation/artwork previews. When they are present, the library reports that limit, so its picture can differ from the effective game view. A blank thumbnail or a missing AnimData frame is not evidence that an ID is free to replace. Use the existing allocation and reviewed-save tools for that decision.

Playback controls

Sequences load paused. Play/Pause starts or stops the loaded sequence. Previous and Next step frames. The slider scrubs the sequence and the frame label identifies the current position. Loop repeats it. At a narrow width, controls wrap onto further rows; scroll the viewer to reach them. Play and image-export buttons stay disabled until a valid sequence loads.

Preview FPS and Speed combine into the effective preview rate. Changing them affects Studio playback and the timing of a saved GIF, not the native movement speed of a creature in game. Inspect both a normal-speed loop and individual frames for alignment problems.

Export choices

Action	Result
Export frame	A PNG of the exact frame displayed before the save dialog opens
Save GIF...	The selected body action/direction or item/effect sequence as an animated GIF, retaining the chosen FPS, Speed, Zoom and Loop settings
Preview in Test Scene	For animated item/effect artwork, opens an optional scene preview of the loaded sequence
Save entire body as .vd…	Saves the selected native MUL body, or a supported routed UOP Animal VD plus a complete source-preservation ZIP; unavailable for a raw UOP library view
Convert routed body to .vd…	For a raw UOP selection, converts the client-routed layout, which can choose different groups from the raw preview
Convert frames / GIF to .vd...	Opens the animation workshop for direct native VD conversion
More export options… / Routed body export options…	Opens bulk export with the body selected; the routed UOP label identifies use of client action mappings
Import Animation…	Choose animated items, spell/visual effects, body/character animations or mounts; then preview and review the matching import
Right-click > Show in Test Scene / Show routed body in Test Scene	Opens the local body scene; a raw UOP selection explicitly labels its routed interpretation

PNG/GIF exports are useful for image editing and demonstration. Saving pauses playback and captures the displayed frame or selected sequence before the file dialog opens. Cancel leaves no export file and keeps playback paused. A GIF of a body contains the selected action and direction; it is not a whole-body package. A GIF of an item/effect does not install its native artwork or AnimData. Save entire body as .vd saves native body data or the supported UOP conversion without first creating PNG folders. Keep body.json/frames.json metadata with bulk-exported folders when you intend to edit and reimport them.

Save a whole body for distribution

1. Open Animations using the sidebar or Ctrl+K.
2. For a routed body, enter its logical ID in decimal or 0x hexadecimal. Selected creature fills it from the selected CHARDEF when available. A MUL library selection can instead be saved directly using its displayed raw bank/body.
3. Keep the displayed raw Source for a direct MUL library selection. For a manually entered logical body, set Source to Auto and click Load body; an alias can resolve to another physical body or bank. A raw UOP library selection keeps its UOP source and offers the separate routed-conversion command described below.
4. Click Save entire body as .vd… and choose the filename and output folder. If you began with a raw UOP library selection, use its explicitly labelled Convert routed body to .vd… command instead; it converts the client layout rather than promising a VD of the raw group list.

5. Read the result and coverage report. A native MUL export includes every available action and stored facing for that body, regardless of the current preview. For supported UOP ANIMAL bodies, the ordinary VD contains the standard Animal set; the companion complete ZIP retains the other source groups and any available riding set. Missing native actions or facings stay empty and are reported. UOP conversion requires visible data in all five stored facings of every source group; an incomplete source produces a clear error before files are saved.
6. Share the saved .vd for its matching ordinary import. Share the complete ZIP as well when distributing a converted UOP body with all its source/riding animations. Recipients should follow the bundle README and review their own destination IDs.

For example, enter body 1653 for the Fire Elemental Horse if it exists in your linked client. Keep Source set to Auto: that body can be supplied by modern UOP archives and have no physical records at the same number in any MUL bank. Searching only All MUL banks would then export no files.

For a native MUL body, Studio retains the original payloads, palette, frame origins, index extra values and full 13-, 22- or 35-action table. Empty source slots stay empty. A VD stores five facings per action; UO mirrors the other three to display all eight directions. Separate mirrored preview frames are not needed in the VD.

What a converted UOP body includes

Supported modern UOP ANIMAL bodies use their own AnimationSequence mode/action tables to prepare the standard 13-action Animal VD. Studio reads the last valid recognized definition for that body in mobtypes.txt, matching the routed viewer; an older duplicate entry does not override a later valid Animal entry. The additional Animal flag 0x2000 is supported alongside the UOP flag 0x10000, including combined flags 0x12000. Other or custom native layouts produce an explicit unsupported-layout error. Representable frames retain their native RGB555 colours, transparency and origins; conversion does not promise arbitrary colours or layouts that VD cannot represent.

The converter also writes a companion complete ZIP beside the chosen VD. It chooses a collision-free bundle name so another existing bundle is not overwritten. The bundle contains:

- The standard 13-action Animal VD for ordinary Animal import.
- A separate 13-action riding VD when the body's AnimationSequence table supplies an available riding mode.
- Archival 35-action-layout VD files preserving all original physical UOP groups, split into files when needed.
- Mapping JSON and a public README explaining the source modes, action assignments, coverage and use of each file.

The archival files use 35-action tables as containers for preservation. They are not normal Animal imports, and their table size does not turn the source into a Human or equipment body. Keep them with the mapping JSON if you need to recover or deliberately remap the original UOP groups.

Some sources supply only a generic fidget for the native Eat/Fidget slots. The standard VD reuses that source sequence where its table calls for it; it does not create a distinct eating animation. A second death slot reuses the first pose only when the source table explicitly aliases that second action to the same pose. Absent native Animal slots 4 and 11 remain empty and are reported. Physical source groups are preserved even when no riding mode is defined, but their presence alone does not create a riding VD. UOP conversion checks every source group and rejects a completely blank stored facing before saving. Native MUL exports retain empty source slots and report missing coverage; mirroring provides display directions only when the corresponding stored source facing exists.

A VD contains animation data, not a server definition, sounds, inventory icon or riding-memory assignment. Keep those related assets and instructions with a distribution when needed. Check standing, walking, running and riding actions in the recipient's matching client/server setup; export does not create absent gameplay poses or timing fields.

16. Animation workshop and layered preview

Edit frames

Open Animation workshop & VD converter from Home, Import artwork or All tools. Load displayed sequence takes the current Animations view. Open exported sequence reads a frames.json-based sequence. Open frame images reads supported PNG/BMP frames or one GIF. Open sequence ZIP lists supported packaged sequences, preferring frames.json exports when present. Choose one source; exported sequence metadata retains its action, stored direction and frame origins. Otherwise choose a GIF or one same-folder image sequence. Preview/reference folders and matching BMP alternatives to PNG frames are skipped. The workshop does not merge unrelated folders or generate missing movements.

Choose the matching VD layout: Monster (22 actions), Animal (13 actions) or Human / equipment (35 actions). Use Advanced to set destination bank, body, action and stored direction for an eventual MUL import. These fields identify the write destination; they do not themselves change the currently loaded source files.

Use the slider to choose a frame. In Advanced, Origin X/Y set the registration point. Apply origin to frame changes one frame; Apply origin to all frames changes the sequence. The ground-origin guides help identify unwanted shifts. Show neighbouring frames provides an onion-skin comparison.

Advanced also provides Move frame earlier/later, Crop transparent border, Replace this frame and Undo. Cropping maintains the intended registration. Play / Pause checks motion at Preview FPS.

Export edited sequence preserves an editable image/metadata folder. Do this before closing unfinished work; general session restoration does not include the workshop's unsaved frame canvas. Review import to MUL prepares the native sequence write. Allow replacing existing sequences must be enabled for an occupied destination.

Convert frames or a GIF directly to VD

1. Open Animation workshop & VD converter from Home, Import artwork or All tools. Animations also offers Convert frames / GIF to .vd.
2. Choose Open frame images and select PNG/BMP frames in natural filename order, or choose one animated GIF. Open exported sequence reads a saved frames.json sequence instead. Open sequence ZIP can load one of those supported packaged sources; choose the intended entry when several are listed.
3. Choose the matching VD layout and inspect the frames. Fresh images use action 0 and stored direction 0; a displayed or exported sequence retains its loaded action/direction. Use Advanced to choose another action, a stored direction from 0 through 4, or manual origins.
4. Choose Save sequence as .vd, select a destination filename and read the completion status.

The result is a native UOFiddler animation VD containing the loaded sequence in one action and facing. All other action/direction slots are empty. It does not supply a complete creature, mount or equipment animation set. Supply and check the required standing, walking, running, riding and other actions before using such content in game.

PNG/GIF images are converted to the native palette: up to 256 colours, RGB555 colour precision and binary transparency. Alpha values below 128 become transparent; values from 128 upward become opaque. Frame origins are retained or taken from your alignment settings. GIF timing and Preview FPS are not stored as native VD animation timing.

This export creates the chosen VD file without writing to a client or server. Studio reads back and verifies the staged native file before replacing an existing destination. Invalid input leaves that existing destination unchanged. Import the result separately through the appropriate reviewed importer when you are ready.

Convert an exported body folder to VD

Choose Convert body folder to .vd and select the folder containing body.json and Action_*/Direction_*/frames.json with their PNGs. For a packaged export, choose Convert body ZIP to .vd and select the intended exported body. Studio uses the declared 13-, 22- or 35-action layout and preserves supported origins and sequence metadata while converting the images. If the layout is absent, switch to Advanced and choose the correct VD layout before trying again.

Only the supplied stored sequences are included. Mirrored preview folders are skipped, and missing actions or facings remain empty. The completion message reports the sequence and frame counts; it is not a completeness check for gameplay. Keep the original PNG folders and metadata for later editing.

A body marked as modern UOP is not converted automatically: its action IDs need deliberate mapping to a supported MUL layout. Import or convert individual supported sequences with explicit action choices when necessary.

Reimport a body folder

Export the body first using the bulk/body exporter. Edit the PNGs while keeping the folder structure and metadata. Reimport body folder reads the body.json, Action and Direction arrangement and preserves supported origins/index information. Reimport body ZIP reads the same supported structure from an archive; select the intended body when several are found. Nested wrappers are discovered, and mirrored display exports remain separate from stored directions.

For a new body, Simple uses automatic free-block selection in the supported destination layout. In Advanced, keep Body folder: use next free block enabled and choose a compatible bank. For an update, switch off automatic body choice, enter the intended destination and enable replacing existing sequences. Review the complete list before saving. A dropped body folder opens the workshop and still requires review.

Mirrored directions exported for viewing are not reimported as new stored directions. A body exported from modern UOP needs deliberate action mapping; this feature is not automatic complete UOP-to-MUL conversion. Use individual supported sequences with explicit destinations if necessary.

Equipment / mount preview

Open All tools > Equipment & mount preview to combine multiple saved animation layers. Add the mount first, then rider/body and equipment. Each layer has its own Body / equipment ID, source, action, direction, hue and X/Y offsets.

1. Enter a body and its source/action/direction, then Add layer.
2. Add the rider or other body using the appropriate pose.
3. Select a wearable in Item artwork and choose Add selected item, or enter its equipment animation ID manually.
4. Select rows to Remove layer, Move layer earlier, or Apply offsets / hue.
5. Play / Pause to inspect the combined motion.

The layers share a ground origin, but you control their order and poses. Add selected item reads the item's TileData animation link. It does not import a new wearable. Offsets, hues and layer order are preview settings

and are not written to native assets or Sphere scripts. This is useful for checking overlap, not a full implementation of the game renderer's equipment rules.

17. Gump artwork: browse, recolour, replace and import

The Gumps tab edits interface artwork records. These images can be backgrounds, buttons or paperdoll pieces. The Sphere gump designer in the next chapter arranges such artwork into a dialog; the two tools have different purposes.

Browse and use a gump

Choose Auto, MUL or UOP and Reload. Search gump ID/name and use All, Free candidates or Used. Scroll thumbnail cards and select one for the full preview. The preview offers Fit and fixed zoom levels.

Enter Preview hue ID and Apply, or Choose hue. Hue 0 shows original colours. Export PNG saves the displayed preview, including its selected hue. Use hue 0 if you want an uncoloured source export. Copy Sphere GUMPPIC line gives a starting command for script editing; position and surrounding dialog logic still need to suit your layout.

Right-click a gump preview/card for image replacement, references, favourite actions and export where offered. Inspect equipment-linked gump IDs before replacing them: a paperdoll slot may be referenced through an equipment animation even if it is not named in a simple dialog script.

Import a batch

Choose Import Gumps on Home, Import artwork or All tools, or click Import PNG / BMP in the Gumps view. Choose the images and destination format. Choose automatic consecutive IDs or deliberate IDs from filenames; review their order and proposed destinations. Browse slots / choose a free range helps find candidates.

Enable Replace the listed gump slots only for intended replacements. That option can include referenced IDs, so review their usage. Inspect the converted image and open the save review, then queue or save. This changes gump artwork; it does not automatically add a Sphere dialog that displays it.

Replace one existing image

The common save review lists existing gump destinations and ownership reasons. Inspect the replacement preview and graphics review details, then give the required fresh confirmation. Existing paperdoll references can make a blank-looking gump protected; queued or unavailable destinations stay blocked.

For an editor round trip, select the image and choose **Edit original artwork**. Studio opens a separate PNG copy in your chosen installed editor, watches for a saved return, and sends it through the same reviewed replacement process. Preview hues are not baked into that original copy. Keep the sidecar metadata unchanged and save the same PNG filename.

Select the correct gump and use its right-click Replace ... with image action. Choose the PNG/BMP and compare the old and converted new image. Keeping the ID means existing scripts/paperdoll links can continue to refer to it, but changed dimensions or alignment can affect every place it is used. Preserve the full intended canvas for paperdoll pieces.

If a gump looks absent or unchanged after saving, verify the active source format, the client's actual data folder, the ID used by the script and any overriding files. Reload the preview after disk changes.

18. Sphere gump designer

Open All tools > Sphere gump designer to arrange a dialog visually and generate Sphere source. It uses existing gump graphics from your client. Import new graphics in Gumps first when required.

Create a dialog

On the Dialog tab, set Dialog DEFNAME to a unique name beginning `d_`, such as `d_example_menu`. Choose Canvas width/height and apply the dialog settings. Supported canvas dimensions are 100 to 4096; the design supports up to 1000 controls. Above the editor canvas, choose Page and Show page to inspect a page; page 0 is shared content.

Choose a control type, fill its properties and click Add control. Click/drag a control on the canvas to move it. Apply properties updates the selected control after editing fields. Layers offers Delete, and Undo/Redo manage layout edits. Property labels identify Normal gump ID, Pressed / checked gump ID, Reply / input / switch ID, Text / input value and Text hue.

Control	Main properties	Generated purpose
Image	x, y, graphic	GUMPPIC artwork
Tiled background	x, y, width, height, graphic	Repeated gump artwork over an area
Text	x, y, hue, text	A text label
Button	x, y, graphic, pressed, id	Reply button with normal/pressed graphics
Page button	x, y, graphic, pressed, destination page	Local page navigation without a reply handler
Input	x, y, width, height, hue, id, text	A text-entry field
Checkbox	x, y, graphic, pressed, switch ID, Start checked	An independent switch
Radio	x, y, graphic, pressed, switch ID, Radio group, Start checked	One selected switch per group

Use the artwork picker to browse existing native gump thumbnails and choose a normal or pressed graphic. Classic Image, Button, Page button, Checkbox and Radio graphics retain their native dimensions. Tiled background and Input support width/height resizing.

Layers and precise layout

The Layers list lets you select, reorder, hide or lock controls. Hidden controls are omitted from the generated dialog; locks preserve visibility and prevent movement or editing until unlocked. Use Ctrl-click in Layers, Shift-click on the canvas, or drag an empty-canvas selection rectangle for multiple selection. Then align, duplicate or delete the selected controls. Snap can keep moves on the selected grid. Undo and Redo restore the complete editable document, including property and layer changes. A drag is one undo step.

Test the dialog locally

Choose **Test / Interact** to use entries, checkboxes, radio groups and buttons in a separate local test window. Page buttons change the visible page; reply buttons show their reply ID, input values and checked switch IDs, retained across pages. This helps check navigation and control IDs before export. It does not run Sphere handlers or reproduce the game client's font metrics. Verify the generated dialog on the receiving server and client.

Page assigns a control to a dialog page. Preview page chooses what you inspect; page 0 is shared content. Text is single-line. Buttons require distinct nonzero reply IDs; inputs require distinct input IDs. A reply button is not automatically a next-page navigation button.

Save the editable design

Save design writes JSON for future editing. Open design loads it. Save this separately from the generated .scp; editing the generated script does not update the JSON layout. General session recovery is not a substitute for saving the designer canvas.

Opening a version 1 design migrates it to the current version 2 model. Saving writes version 2. Keep an original copy if you need to open it in an older Studio.

View, export or install its script

View script shows the generated DIALOG and button-handler sections. Export script writes that text to a chosen output file. That export does not register the file with your server.

Install script prepares a reviewed new .scp inside the active server's scripts directory and a resource entry when needed. Choose a unique filename and dialog DEFNAME; existing dialogs are not silently replaced. Queue or save the reviewed operation, then open the saved script for custom behaviour.

Generated button handlers are stubs: they include a comment for your Sphere action and a return. You must implement what each button does. The desktop preview also uses desktop font metrics and cannot verify the game's exact text layout, input behaviour or your server-side handlers. Test the dialog in the target client after scripting it.

19. Hues and colour testing

Open Hues to browse the client's colour tables. Reload refreshes them. Search by name/ID, choose a hue and compare its 32-swatch palette with the image preview.

Game hue 0 means no colour transformation. File palette index 0 corresponds to game hue 1. Use the game hue ID shown by Studio when copying a Sphere COLOR assignment.

Preview a saved hue

Test image shows a standard sample. Load PNG uses your image. Selected item and Selected gump use available assets from those views. Preview grey pixels only limits the comparison to grey pixels; it does not change the item's saved PartialHue flag.

Copy Sphere COLOR copies a script assignment. Export preview PNG saves the coloured preview. Export hue list CSV exports the hue listing, not a replacement hues.mul. Import palette / Export palette use the supported UOFiddler palette-text interchange.

Edit or copy a palette

Select a hue and Edit hue. Change its short name, select one of the 32 swatches and use Set RGB hex or Colour picker. Create gradient fills the palette between chosen dark/light colours. Import UOFiddler palette can populate it from a file.

The preview uses the unsaved working colours. Test this unsaved hue on clothing / items opens a sample comparison without saving it first. Review hue changes shows the proposed write to that hue record; then queue/save.

Copy to slot prepares another palette record from the current one. Choose the destination carefully: a blank table is only a candidate and may still be used by world objects or scripts. Copying a palette does not automatically update the COLOR of items that used the source hue.

Stored fields includes TableStart and TableEnd. These are stored 16-bit fields, not the indices of the first/last swatch you want to edit. Preserve them unless you have a specific reason to change or import them. The reviewed write updates the selected hue record, retaining other records and block information.

Compare clothing and items

Try clothing / items lets you select up to 24 items by searching Sphere names/categories or using Item IDs / range. Choose Item art, Male clothing or Female clothing. Choose Item flags, Full hue or Grey pixels only to control the comparison.

The clothing modes show individual paperdoll pieces, not a complete animated dressed character. Choose hue switches the palette. Export comparison PNG saves the resulting board. Check more than one material: a hue that reads well on a grey robe may be poor on an already coloured weapon.

20. Sound workshop and attachments

Open All tools > Sound workshop. Select Auto, MUL or UOP, then Reload. Search the table by ID or name. Selecting a sound shows its waveform and duration. Play and Stop audition it through Windows audio.

Export sounds

Export WAV saves the selected sound with a WAV header. Export all WAV writes the available sounds to a new output folder. A sound archive's internal bytes are not themselves a complete WAV file; use the exporter rather than renaming raw archive data.

Import or replace audio

The WAV import processing settings apply to the files you subsequently choose:

Setting	Meaning
Trim start	Start position in seconds
Trim end	End position in seconds; blank keeps the end of the source
Volume multiplier	1 keeps level; lower reduces volume; higher may clip
Fade in/out	Fade duration in milliseconds

Choose Add WAV files for new candidate IDs, or Replace selected for one existing sound. New allocation considers supported archive entries, mappings/overrides, literal sound references and pending reservations. A replacement affects existing uses of that sound ID.

The importer accepts uncompressed PCM WAV at supported mono/stereo 8/16/24/32-bit depths. It converts to the native signed 16-bit mono, 22050 Hz sound payload. MP3 and floating-point WAV need external conversion to supported PCM first. Resampling is simple; prepare demanding audio work in a dedicated editor.

Inspect the duration and clipped-sample count in review. Play converted sound auditions the prepared payload; Stop stops it. With a batch, the review's audio audition is the first prepared sound, so inspect individual sources/results when validating a collection. Queue/save only after checking destination IDs and conversion.

Shorten a source filename if the derived sound name exceeds 39 Windows-1252 bytes or contains unsupported characters. Renaming does not change the sound's gameplay assignment.

Attach a sound to an item, creature or mount

Use this workflow to connect audio to an existing definition. Finish or discard open script drafts first. Scripts in a server ZIP are read-only; extract and link the server before attaching sounds.

1. Select the item, wearable, creature or mount in the definitions list.
2. Open Workspace and choose Attach sound to this asset... in the Sound in UO panel. Right-click > Attach sound... on a definition, or All tools > Attach a sound, also opens this window.
3. Check Attach to. It identifies the exact script section and line; other supported sections in that script are available in the list.
4. Choose Play when using the choices below.

5. Choose Auto, MUL or UOP for the saved sound source. Search by name, decimal ID or `0x` hexadecimal ID, then select a sound.
6. Press Play new sound or double-click the sound to hear it. Stop ends playback. The waveform and duration help you inspect the selected audio.
7. Choose Review sound attachment.... Read the sound ID, target, event and before/after script text. Listen to the review's audio if needed.
8. Choose Queue changes to defer the reviewed operation, or Save to UO files to save it now. Enable Keep a backup if you want retained originals for later restoration.

Save to UO files writes to the client-data and server-script folders linked by the active profile, after the safety checks. You do not have to paste the sound fields into the script yourself. It does not upload to a website or reload a running server; restart or reload the intended client/server after saving so the sound is used in game. For a new WAV, players also need the updated sound archive in the client data they actually use.

Target	Play when	Script connection
Creature or mount	Idle	SOUNDIDLE
Creature or mount	Notice / alert	SOUNDNOTICE
Creature or mount	Attack / hit	SOUNDHIT
Creature or mount	Get hurt	SOUNDGETHIT
Creature or mount	Death	SOUNDDIE
Item or wearable	Use / double-click	@DClick
Item or wearable	Equip	@Equip
Item or wearable	Unequip	@UnEquip
Effect / function	When this function runs	Start of the selected FUNCTION

A saved sound is reused without rewriting its archive. The definition or function supplies the gameplay connection: attaching to an item does not make it equipable, and attaching to a mount does not add vehicle physics or engine timing. Sound 0 is not offered because it represents no sound in Sphere.

You can also select a sound in Sound workshop and choose Attach selected sound to a thing..., or right-click the sound and choose Attach sound.... Select the intended definition, or open its script, before using that route. Check Attach to again before review.

See and replace sounds already associated with an asset

Open Attach sound and select Current sounds. The list shows the event, saved sound ID and name when available, and the script location supplying the link. Creature rows include sound fields inherited through a resolvable ID and the action sounds derived from a base SOUND set. Expressions and special server-chosen sounds remain labelled instead of being guessed.

1. Select the current sound row. Play current sound auditions that saved audio without changing the replacement you have selected.
2. Click Choose replacement. The Change control selects Set / replace selected sound and the Choose new sound tab opens.

3. Select another saved sound, or choose a ready-made WAV. Play new sound lets you compare the replacement with the current sound.
4. Choose Review sound attachment. Check the old-to-new value and the exact script change.
5. Choose Save to UO files or Queue changes.

Replacing an item or function sound updates only the selected command, keeping its object prefix, conditions, other arguments, comments and position. Other sound commands remain. Replacing an inherited creature action creates a local override for that creature; the parent definition stays unchanged. If the local field appears before ID, Studio moves the override after that inheritance copy so it takes effect.

Replacing a link does not overwrite the old shared audio slot. A WAV receives a free slot and is saved together with the changed link. Other assets that use the old sound keep it. To deliberately change shared audio everywhere, use Sound workshop > Replace selected instead and review all affected uses.

For an item or function, Add a sound preserves the existing commands and inserts or updates Studio's marked attachment at the selected event's start. Use Set / replace selected sound when you want to swap an existing command. External event scripts, called functions, runtime property changes and dynamic expressions are not fully evaluated here. An unresolved entry does not prove that the asset is silent; inspect its script or verify the event in game. Client sound.def or loose-file overrides can also affect playback.

Attach a ready-made WAV in one save

In Attach sound, choose Choose WAV file... instead of selecting a saved sound. Select an uncompressed PCM WAV, then use Play new sound to hear the converted audio. The supported input formats and filename limits are the same as the sound importer above. The attachment window converts the file without offering the workshop's trim, volume and fade controls; prepare those edits beforehand if needed.

Studio chooses an available, unreserved sound slot for the WAV. Review shows that new ID and the script connection together. The sound data and script edit are saved or queued as one recoverable operation. Existing sounds and pending reservations are considered when choosing the destination. If no suitable slot is available, reuse a saved sound or choose another compatible archive format.

Selecting a saved sound after choosing a WAV switches the attachment back to reusing that saved sound. Check the source note and review to confirm which audio will be used.

Attach a sound to an effect function

1. Open the effect's .scp from the Script directory so the whole file is available.
2. Choose All tools > Attach a sound.
3. Under Attach to, select the intended Effect / function entry. A script can contain several functions; use the identifier and line to choose the correct one.
4. Choose When this function runs, select a saved sound or WAV, and review the attachment.
5. Save or queue it, then verify the event with the target client and Sphere after their data is reloaded.

The inserted sound runs at the start of that function and uses the object calling it as its source. A function invoked without a world object may need a manually chosen sound source in its script. An artwork ID or animation frame alone does not execute audio; merely placing an effect item does not call a separate effect function.

Test Scene remains a visual preview. It does not execute Sphere events or automatically play attached sounds. Use Play new sound in the attachment window to audition the audio; verify event timing in the target game setup.

Existing scripts, inheritance and repeat attachments

For a creature, Studio sets only the selected sound field on that definition. Other fields, inherited links and triggers remain intact. A local sound value overrides the corresponding inherited value without rewriting the parent definition.

With Add a sound for an item event or function, Studio adds a marked SOUND command before existing conditions and returns. Existing custom logic and other sound commands remain. This can play audio even if a later condition rejects the action, or produce more than one sound if the script already plays audio. If the sound should run only after a successful action, deliberately move the command to that point in Sphere scripts and review the logic. Set / replace selected sound keeps the chosen command at its current position.

Using Add a sound again on the same event updates Studio's unchanged marked command instead of adding another copy. That insertion mode requires manual review of duplicate triggers or edited/moved markers. To replace an existing command, select its exact row in Current sounds; Studio does not choose a custom command for you.

If the script already has a queued edit, save that edit or remove it from the queue, then reopen Attach sound. A saved sound with a queued replacement also needs to be resolved first. On-disk changes invalidate the attachment review; reload the current files rather than overwriting newer work. To add several sounds to the same script, finish each save before reopening the next attachment.

The browser previews the selected saved MUL/UOP archive. A game client's sound.def mapping or loose-file override can change what actually plays. Music is outside the sound workshop; verify the intended event, source and volume in the target client/server.

Copy creature sound fields manually

Creature sound fields opens SOUNDIDLE, SOUNDNOTICE, SOUNDHIT, SOUNDGETHIT and SOUNDDIE entries. Enter the intended sound IDs and Copy Sphere fields. Paste the assignments into the appropriate CHARDEF and save the script separately.

This helper remains available for manual script work. For a reviewed direct attachment, use Attach sound... above. Importing audio alone does not assign it to every creature automatically.

21. Test Scene: an in-app visual sandbox

Open the Test Scene tab, All tools > Test Scene, or Preview in Test Scene beside the definition list. Right-click item art, an Asset spaces card or the loaded animation preview for Show in Test Scene. A raw UOP library preview labels its body-scene command Show routed body in Test Scene because client action aliases can select different groups.

For native item/effect playback, open Animations > All animations > Animated items & effects and double-click a result. The sequence plays directly in Animations, where you can also save a GIF or PNG frame. Choose Preview in Test Scene to open it here as Kind: Effect with Effect frames: 0. This optional scene route uses the selected client's AnimData frame order; it does not require an item or creature definition. Chapter 15 explains the library and its source notices.

The scene runs entirely inside Studio. It can show characters, mounts, items, equipment and effects using saved files, supported queued edits or an unimported preview file. Its movement and obstacles are local; Sphere combat, ownership, taming, feeding and script triggers are not executed.

Attached sounds are not played automatically in this scene. Audition them in Attach sound or Sound workshop; check their actual gameplay event using the target client and server.

Load content

Load selected definition follows the current selection. Alternatively choose Kind, enter its ID and Source, then Load / refresh preview. Character/Mount IDs are animation bodies; Item/Equipment/Effect use their relevant item-art links. Read the source note above the scene to see what was loaded.

Include queued changes overlays supported reviewed animation, item-art, effect and equipment metadata/paperdoll changes in memory. Enter the proposed destination ID when testing a queued import. Refresh after changing the queue. This preview does not commit the queue to disk.

Preview VD / PNG / GIF opens supported source files before importing. Return to saved / queued files stops using that external preview file. A source-file preview cannot invent missing directions or actions.

Character and mount controls

Choose Idle, Walk, Run or Custom pose. Custom uses the Action ID. Direction selects one of eight displayed directions. Player body selects the reference/rider body where that control applies. Hue changes the preview colour; FPS and Speed change playback.

Mount mode offers Mounted rider and Rider X/Y offsets. Show reference player adds an optional comparison player in other suitable modes. Equipment mode uses the item's equipped animation and retains the player needed to display it. Missing required player art may be represented by a labelled placeholder.

Play/Pause controls animation. The frame buttons pause and step. Click the floor or focus the canvas and use arrow keys to move. Space toggles playback. These controls affect the preview, not native frame data or server speed.

Place an item without replacing the character

Selecting an item for Show in Test Scene prepares ground placement while retaining the character/mount already in the scene. In Room, choose Place item and click floor tiles to place copies. Use Move character when you want clicks to move the actor again. Escape returns to movement.

Wear in Test Scene, available on the small paperdoll, is the clearer route for an equippable item: it shows the item on the human model with the corresponding paperdoll. Ground placement and wearing are different actions.

Effects

Effect frames 0 uses AnimData when available. A positive count previews explicit consecutive art frames. Effect style selects Ground, On player or a looping Projectile demonstration. Check hue, scale, frame order and overlap. This does not verify the generated Sphere effect function's damage or timing rules.

Library item/effect selections start a fresh ground preview, clearing a previous body/file/rider preview. They use the selected sequence's signed/repeated AnimData offsets, not a guessed consecutive range. FPS and Speed control this demonstration; native interval/start bytes shown in the library are not converted into promised game timing. Missing artwork frames stop the preview with an explanation. Item/effect reading does not require unrelated body-routing metadata, although adding a reference player still requires its body assets.

Room and wall controls

Control	Use
Scene tool	Move character, Place item, Add wall or Remove placed
Wall direction	Choose Left or Right orientation for added walls
Lighting	Day, Dusk or Night preview brightness
Room	Client stone, Plain grid or Custom environment
Walls / local collision	Toggle the room walls/local obstruction behaviour
Floor grid	Show the floor tile grid
Place selected item	Prepare the selected item's art as a prop
Selected item as floor / wall	Use selected art for that environment role
Reset room	Clear the local arrangement and reset the room

The default walls follow the two back edges. Add wall lets you place test obstacles by clicking tiles; Remove placed clears placed items/test walls at the clicked tile. Environment art uses available client pieces, with a plain-grid fallback. Floor art is fitted to the scene's ground diamonds.

Zoom, snapshots and persistence

Animation playback and Test Scene reuse prepared preview images within bounded memory budgets. Relevant source, project and pending-edit changes refresh the previews. This improves repeated drawing without changing exported pixels or the preview's rendering order.

Auto zoom fits smaller windows and limits enlargement; Fit fills the available space. Fixed zoom choices provide closer inspection. Snapshot exports the rendered image as PNG.

The scene arrangement, movement, rider offsets and lighting do not change scripts, game files or queued imports. A snapshot is a demonstration image, not a saved playable world or editable house. Use House Builder when you want a design that can be compiled into a house/deed.

22. House Builder and housing deeds

Open All tools > House Builder. This is a separate in-app editor for a fixed house assembled from existing item artwork. It saves editable .house.json designs and can compile a native multi with a matching Sphere house and placement deed.

Start a design

In House settings, set House name, Plot width/depth, Storage limit and Deed value. Apply plot settings applies validated changes. Plot sizes range from 3 by 3 to 31 by 31; piece heights range from -20 to 100. A design can contain up to 10000 pieces.

Choose Building pieces to search native art by name or ID and filter by role. The material selector offers matching groups such as timber, stone, brick or logs with associated roof/floor/door pieces, plus All building pieces, Favourites and Recent. Available results depend on the client's artwork. The star button toggles the chosen art in favourites.

Building kits, stamps and Problems

The building-piece browser can read supported building tables from your selected client and group pieces into kits and roles. Choose a **Client kit** for Walls, Doors, Floors, Roofs or Stairs, then search for the desired variant. Wall and roof tables can identify north, east or corner pieces; doors and floors retain numbered variants where the source table has no orientation information. If the tables are absent or incomplete, **All building pieces** and custom artwork IDs remain available. Check the role of each piece before compilation.

Select a useful group of pieces, choose **Stamps** and **Save selected**, then give it a name. Choose **Place stamp**, move its ghost preview to the intended location, and click to place it. A placement is one undo step. Import and Export share `.house-stamp.json` files containing artwork IDs, relative positions and roles. The saved library is scoped to your active project beside Studio settings. Stamps do not install into the client or server by themselves; share matching artwork separately.

Check building opens the Selection panel's **Problems** list. Select a finding to inspect its location or associated piece; some general findings have no piece target. Optional automatic checking waits briefly after a design change; moving the pointer does not trigger a new check. Findings describe the current design and are cleared when stale. The existing compile review and walk test remain available.

Pick the correct role

Artwork ID selects the image; Role tells the builder what that piece does in the design. Inferred roles are a starting point, so check them. Floor, wall, roof, stairs and decoration are static pieces. Door and sign are functional components for the compiled house; the selected native art must support that function.

Every compiled design needs floor content, at least one working door and exactly one house sign. The interface suggests a standard brass sign, but the selected server must actually have a compatible inherited sign definition. A sign-like image assigned Decoration is not a house sign.

Drawing and editing tools

Tool/control	Action
Paint	Click or drag to add the chosen art at the active height
Line	Drag between endpoints to draw a line of pieces
Rectangle	Drag to fill a rectangular area
Perimeter	Drag to draw the rectangle's boundary
Erase	Remove the hit piece; right-click also erases a hit piece
Select	Click a piece, or drag a selection area at the active height
Eyedropper	Pick a piece's art, role and height
Copy / Paste	Copy selected pieces, then place their relative arrangement
Delete	Remove selected pieces from the design
X-/X+/Y-/Y+	Move selected pieces one tile in the chosen axis
Lower / Raise	Move selected pieces one height unit
Undo / Redo	Step through design changes

The translucent ghost shows prospective placement, including copied groups. Cursor text gives tile X/Y and height. An out-of-bounds paste preview is marked differently; the design still has to pass validation. Large previews may show only an initial portion of the placement group, so review the finished selection too.

Floors, visibility and selected-piece properties

Height chooses the active building plane. Presets include Ground 0, Main floor 7, First floor 27, Second floor 47 and Roof 67. The -20/+20 buttons move the plane quickly. These are convenient heights, not an automatic multi-storey generator.

Roofs toggles roof visibility. Upper floors controls whether pieces above the active height are shown. Use Fit or a fixed zoom, middle-drag to pan, and Centre view to reset pan and fit the plot.

Open Selection after selecting pieces. Artwork ID, X, Y, Height and Role describe the selection. For a group, coordinates reposition the first piece while preserving the other pieces' relative arrangement. Blank artwork or role preserves each piece's value. Apply to selected pieces validates the result and records an undoable change.

Save, open and recover editable work

Save project writes .house.json; subsequent saves reuse its path. House settings > Save project as creates another editable file. New and Open project handle unsaved work before replacing the current design. Import blueprint reads supported unhued CentrED CSV into the current project; check inferred roles afterward. Export blueprint writes the supported tile data as CSV.

CSV is interchange data, not a fully registered Sphere house. It does not carry the project's ownership link or every editor setting. Per-piece hues and unsupported visibility flags are rejected for fixed-house import. Share a .house.json when you want to preserve the editable roles and settings.

House Builder writes automatic recovery copies while you work. House settings > Recover a project opens those copies; then Save project chooses a permanent filename. Recovery is a convenience, not a reason to stop keeping deliberate project files.

Save/Queue in the compile review leaves the builder open. Save the editable project afterward to retain its compiled-house link. Undo and redo preserve that destination link; they do not undo an already completed game-file save. Use reviewed-save recovery for a file rollback.

Check and walk through the house

Check building reports low headroom, unreachable floors/entrances and enclosed floor gaps. Problem pieces are selected where applicable. A compile also presents architectural warnings for review. Some openings may be intentional; read the warnings in context.

Start walk test places a reference character outside the plot. Use arrows/WASD or click a destination for a local route. E or Space opens/closes a nearby door. Step out of its tile before closing it. Open doors appear faded in the preview. Stop walking cancels movement.

The walk test uses native heights, stairs, doors, headroom and blocked-corner checks within a bounded flat-ground simulation. It is not a proof of actual terrain placement or Sphere permissions. Test roof visibility and different floor levels to see what the character is doing.

Compile house + deed

1. Save the editable project and verify its roles, entrances, stairs and sign.
2. Choose Compile house + deed and review any architectural warnings.
3. Read the chosen free multi ID, affected client files and complete Sphere script.
4. Confirm the house name, deed, storage/value and resource registration.
5. Queue or save. When both native multi representations are present, the compiler updates MUL and UOP together.
6. Save the editable project again to retain its link to the compiled house.
7. Load the changes on a compatible test shard, create the generated deed and test placement, entry, ownership and redeeding.

Scripts go into scripts/housing/custom. The selected server must already provide compatible housing systems. Compiling is not a complete replacement for Sphere's housing code, and the editor does not place the house in the live world for you.

Update a compiled house

Open the saved linked .house.json, make changes and compile again. Studio retains the same multi ID, script filename and named house/deed identity, even when the displayed house name changes. The review shows previous and updated script text.

The saved link verifies the destination script and native multi payloads. If they changed externally, belong to another project or no longer match, the update stops. Save queued changes before compiling another update to the same house. Keep the matching project with the target that owns that allocation.

For a house compiled in 0.23.0, open its original unchanged design and use House settings > Link an earlier compiled house. Studio requires an exact script/art match before reviewing the link. Save that link and the editable project before editing further. A changed old house cannot be adopted merely by giving a new design the same name.

Already placed houses retain saved fixtures and region information. Redeed and place a fresh house to apply structural changes. Back up and plan deployment with the shard operator when houses have owners or stored belongings. Studio does not automatically migrate occupied world houses.

Share houses

Share the editable design and a preview image, plus required nonstandard tile assets and setup notes. A linked project targets its recorded compiled house; it is not a universal allocation for another shard. Preserve your linked original. For a fresh destination, create a new unlinked design, exchange the supported blueprint and recheck roles/settings, then compile against that destination's current data.

Generic .uoasset content-pack remapping does not support compiled houses in this version. House Builder is for fixed house designs, not full map editing, a networked CentrED server or the complete custom-housing protocol.

23. Find references and investigate shared content

Asset workspace > Find references and applicable asset right-click menus open a report for the selected asset. Script workspace also offers symbol-oriented reference navigation. Use these tools before changing a shared appearance or deleting old content.

The report lists file, line/record and evidence. Double-click a resolvable script/definition row to jump to it. Export reference report writes the report with its coverage description so you can retain or share the findings after removing private details.

Definition's reference list focuses on supported indexed relationships. The broader report scans script and mapping mentions; item scans also inspect supported unambiguous classic multi layouts and statics. Large scans run in the background and can be cancelled between units of work.

Interpret results carefully

Asset spaces uses indexed references as protection reasons for saved and staged symbols. A Referenced card can be blank. This protection supplements reference browsing; it does not scan world saves, map placements or arbitrary dynamic expressions.

The same number can mean an item, body, coordinate or unrelated value. Read the line instead of deleting every numeric match. A dynamic Sphere expression may use an asset without a simple literal ID. World-save objects, UOP multis, every override and complete runtime behaviour are not universally resolved.

A clean report is therefore not proof that an ID is unused. Combine it with allocation evidence, source inspection and knowledge of the shard's world data. If several definitions intentionally share an animation, choose a new destination for a visual change that should affect only one of them.

24. Remove definitions, animations or whole scripts

Select a definition and use Remove related content, or its right-click equivalent. Choose the removal scope before preparing the final review.

Scope	Intended result
Selected definition only	Remove that definition while keeping animations, art and companions; outside references may remain
Definition and related content	Consider its linked mount/figurine definitions and removable unused assets while protecting shared content
Entire animation group	Include other definitions using the same animation and their linked removable content

Review a removal

1. Choose the intended scope.
2. Read REMOVE and KEEP / REVIEW MANUALLY in the review.
3. Check every listed definition, art/animation slot, mapping and script-file change.
4. Resolve reported external references before requesting a larger removal again.
5. Choose Remove listed content only when the list matches your intent.
6. Refresh the directory/definition view and inspect the result. Keep the automatically retained backup until you have verified it.

Removal is deliberately reviewed and keeps a backup; it is not an unreviewed one-click purge. A larger animation-group scope is useful for genuinely shared custom content, but does not authorise deletion of arbitrary unrelated scripts that mention it.

Why a .scp may remain

A file may contain other definitions, functions or meaningful content. Removing one section does not necessarily delete the whole file. Cleanup can remove a now-empty script and related exact resource references when applicable, but shared content remains.

Open the remaining file and inspect it. Use Remove empty .scp file for genuinely empty content, or Delete script for an intentional reviewed whole-file deletion. Deleting a whole script keeps client assets and other scripts; it can leave references that you must update. The two operations serve different purposes.

What is not automatically removed

Dynamic references and existing world objects are not comprehensively cleaned up. Sounds, gumps and additional effect frames are not all inferred and removed automatically. Shared/ambiguous item uses or static placements may cause assets to be kept. Read the review's actual list rather than assuming “related” means every conceivable dependency.

If Studio reports an outside reference, open the named file/line. Decide whether to update the reference, retain shared content, or choose definition-only removal. Do not erase a whole decoration/worldgen file just to silence a reference warning. Use a retained backup to restore a mistaken completed removal as described in chapter 4.

25. Bulk export and resuming exports

All tools > Bulk export exports saved item art and animation bodies. Animations > More export options opens it with the current body range selected. Pending edits are not included; save them first if they must be part of the export.

Choose export content

Option	Result
Item artwork as PNG	Exports static item images from the selected source
Item source Auto / MUL / UOP / Both	Controls which item representations are exported
Animation bodies	Exports bodies within the specified range
Animation source	All available, all MUL banks, a particular bank, or UOP
Individual PNG frames	Image files for supported action/direction frames
One .vd per MUL body	A complete compatible native body export
Include mirrored directions 5-7	Extra display-direction PNGs, not new stored sequences
Align PNG frames on a shared canvas	Consistent canvas within each action/direction

Set the body range using decimal or 0x values. The range can include bodies without Sphere definitions; empty slots are skipped. A full export may contain hundreds of thousands of files, so begin with a small relevant range when checking settings.

Run and inspect an export

Choose an existing output parent folder and click Export. Studio creates a new organised UO-Export folder. Item images are arranged by source/category/subsection and ID/name. Animations are arranged by source, body type, body, action and direction. Metadata records origins and source IDs beside the images.

Open export folder reveals the result. Check the completion status and error report; an export containing some successful files may still have skipped/unreadable assets. Original archives are not modified.

No files exported means the scan finished without matching any selected artwork, PNG frames or VD files. The reports identify the body range and source and give advice about physical IDs, client aliases and UOP. Report files alone are not an animation export. Use Source Auto in Animations to inspect the resolved source; for a single complete body, choose Save entire body as .vd as described in chapter 15. Completed with errors, Incomplete and Cancelled remain separate results. In a successful VD-only export, zero PNG frames is normal when the VD count is greater than zero.

Keep body.json and frames.json information with their associated PNGs for editing/reimport. Bulk VD export preserves original compatible MUL data; this bulk tool exports modern UOP as PNG without guessing VD mappings. Use the single-body save in chapter 15 for the supported UOP Animal VD and companion complete ZIP.

Cancel and resume

Cancel export requests cancellation between units of work. Retain the incomplete folder if you intend to resume it. All tools > Resume export asks for the incomplete supported export folder and verifies its metadata and completed outputs.

Resumption reuses completed images only after verification. Changed source archives or manually edited exported images can stop resumption. Keep a separate copy for artwork editing rather than editing the folder you still plan to resume. If the inputs have intentionally changed, start a fresh export.

Bulk export is for saved artwork/animation images, not a dependency-complete server content package. Use Content packs for explicitly queued imports and their linked scripts, and House Builder project/blueprint files for editable houses.

26. MUL / UOP conversion and file checks

Convert supported UOP archives

Open All tools > MUL / UOP converter and choose Direction: UOP to MUL. Add UOP files lets you select archives; Add from linked client finds supported archives in the active client's data. Select an unwanted row and Remove selected to leave it out.

Choose an output parent folder, then Convert. Studio creates a new UO-MUL folder with the converted files and a conversion report. Open output folder shows the result. Original UOP files stay unchanged; output is not automatically installed over the active client.

Supported conversion families include item art, gumps, sounds, maps and multis. Multi conversion uses the High Seas format. The original application's modern animation UOP viewer is separate: those animation archives are not converted by this utility.

Check the report and use the matching generated MUL/index pairs. Do not copy only a new index over unrelated data, or assume conversion updates the client's source preference. If both MUL and UOP are present, confirm which one the game and each Studio tool are reading.

Cancel conversion stops between supported work units. Inspect status before using any output; a partially populated output folder is not a completed conversion. Conversion is not map editing, a patching service or automatic client-version migration.

Convert MUL files to UOP

1. Open All tools > MUL / UOP converter and choose Direction: MUL to UOP.
2. Choose Add MUL files... and select the supported data files. Keep each matching index in the same folder; you select the data file, not its index. Add from linked client can find the supported data files for you.
3. Read the listed output filenames and remove any unwanted entries. A batch cannot contain two data files that would produce the same archive.
4. For multi.mul, choose High Seas (16-byte) under MUL multi format only when that is the source's verified format. Keep its original housing.bin and multi-components.txt beside it, as explained below.
5. Browse to an output parent folder separate from the working game installation, then choose Convert.
6. Wait for packing and verification. Studio reads every completed archive entry back and checks its payload before exposing the completed folder.
7. Choose Open output folder. The new UO-UOP folder contains the archives and conversion-report.json; read the report before using the results.

Selected data file	Required companion	Generated archive
art.mul	artidx.mul	artLegacyMUL.uop
gumpart.mul	gumpidx.mul	gumpartLegacyMUL.uop
sound.mul	soundidx.mul	soundLegacyMUL.uop
mapN.mul / mapNx.mul	None	mapNLegacyMUL.uop / mapNxLegacyMUL.uop
multi.mul	multi.idx, housing.bin, multi-components.txt	MultiCollection.uop

Occupied asset IDs and payloads are preserved. Empty index slots are not stored as assets in UOP, and archive compression, offsets and framing can differ from an earlier archive. Art is stored without compression; the other supported families use zlib. The output is verified as a readable archive, not as a byte-identical recreation of an original client's packer.

Only terrain is included when packing maps. Map statics, map differences, scripts, TileData, hues and other separate metadata remain separate. anim.mul and modern AnimationFrame/AnimationSequence conversions are not supported here. Changing the container does not create missing artwork, expand the client's capabilities or establish compatibility with every client build.

Preserve multi housing data

MultiCollection UOP needs information that is not contained in multi.mul alone. Extract the matching original MultiCollection.uop with UOP to MUL first and keep housing.bin and multi-components.txt together with the resulting High Seas multi.mul and multi.idx. The housing file contains the housing catalogue; the sidecar preserves component IDs used by structures and boats. Keep even an empty, header-only sidecar produced by the extractor.

The reverse converter requires both companion files and an explicit High Seas (16-byte) format choice. It rejects unsupported flag bits, mismatched sidecar artwork/positions, duplicate sidecar entries or references to missing tiles instead of discarding them. Older 12-byte multi.mul data is not supported. Do not choose High Seas simply to bypass the format check or invent empty companion files for an unknown source.

Conversion safety and using the result

Both directions check access to the required source files and output folder before creating temporary output. Sources must remain stable throughout conversion; another file-access check and source-revision comparison run before publishing the completed folder. Missing indexes, busy files, damaged records, changed sources or cancellation stop the batch without installing a partial result. Close programs using the reported files and retry when they are stable.

The converter always creates a new result directory. It does not overwrite the selected MULs/UOPs, switch the active profile to the output or install it into a live client. Changing Direction clears the selection and completed-result link; choose the new direction's inputs before converting again. Retain the original complete client data and test the result in an independent copy. Keep a local conversion report private if it includes personal source paths.

Check game files

Tool readiness shows which artwork, gump, animation, colour, sound, building and world workflows have the required checked files. Missing and damaged files are distinguished, and valid alternative MUL/UOP sources are identified. **File details** keeps the complete audit and exportable report. Readiness checks native file structure; it does not guarantee that every contained image, action or server definition is usable.

Item artwork > Check game files is a read-only structural diagnostic. It inspects supported client file arrangements and reports issues such as recognised layout, inconsistent files and whether Sphere's MulFiles setting points to a different location than Studio's client folder.

Save the report when troubleshooting. It does not save pending edits or repair files. Include the exact Studio version, relevant action and the latest associated client/server error when asking for support; remove personal paths or private content from public reports.

If the check finds a mismatch, correct the profile or intended server configuration deliberately, then rerun it. Do not combine indexes/data from different installations or apply wholesale archived files merely because one program starts successfully afterward.

27. Compare projects and create independent test copies

Compare saved files or logical assets

Open All tools > Compare projects. Choose the other folder and the comparison type. Compare with linked client compares against the active client's saved data; Compare with server scripts performs a file comparison for the server source.

Files compares file-level differences and is read-only. Items, Gumps, Sounds and TileData compare supported logical records, so moved archive offsets alone do not necessarily count as changed content. This helps identify actual asset changes after an archive has been rebuilt.

Select a populated record in a supported asset comparison and use Review selected assets from other folder to prepare its import into the active client. Read the review carefully: the other folder is the source; the active profile is the destination. It does not replace a whole archive as a general synchronisation operation.

Export comparison report retains the results. File comparisons do not provide a universal merge tool for scripts, maps or entire client installations. Use the script editor for deliberate source changes and the appropriate importer for supported assets.

Make a test copy

1. Finish pending work and ensure the source files are stable; close programs that keep changing them.
2. Choose All tools > Create test copy.
3. Choose a parent folder outside the existing client and server folders.
4. Enter a new simple folder/profile name.
5. Wait for the verified copy and profile registration.
6. Select that new Shard profile when ready.
7. Read TEST_COPY.txt before launching anything from the copy.

The tool copies server/client data, excluding certain application backup/cache folders and linked-path cases, and checks for changing sources. It updates recognised MulFiles and ScpFiles entries in copied INI files to the copy. It does not rewrite every custom absolute save/log/database path or configure all network settings.

The optional client executable path is still shared as a remembered executable location. Verify its data-directory setup separately. Do not assume launching it automatically uses the copied assets.

A copied server can contain accounts, saves and configuration from the source. Keep the test copy private; it is not a cleaned public distribution or a provisioned virtual server. Review custom external paths before starting Sphere so testing does not write outside the intended copy. Studio does not start the copied server for you.

28. Portable content packs and distribution

Open All tools > Content packs to move explicitly reviewed content between projects. A .uoasset pack contains the pending bundle you chose to prepare, including supported linked scripts/mappings. It is not an automatic export of every dependency used by a selected definition.

Export a content pack

1. Prepare the required imports/edits and Queue changes instead of immediately saving them.
2. Inspect Pending changes and remove unrelated operations from this bundle.
3. Include the required icons, animations, gumps, hues, sounds and script changes deliberately.
4. Open Content packs and choose Export pending bundle.
5. Save the .uoasset outside your working client/server files.
6. Test importing it against a separate suitable project before distribution.

Content packs reject queued appends to existing script files, including a whole armor set added to an existing .scp. This avoids copying unrelated old definitions into the recipient bundle. Prepare the generated scripts with New script for portable distribution; save deliberate existing-file additions through the normal review instead.

Exporting the pack does not save the queued changes to your game files. If you already saved and cleared the queue, the pack exporter cannot reconstruct an entire bundle just from the current definition. Prepare a deliberate supported export/import bundle again or distribute source assets with clear instructions.

Share an animated item or spell effect

Load the ordered frames with Import Animation > Animated items or Spell / visual effects. Check the animation option, native frame interval, preview and Server script choices. Choose Review files and script, then Queue changes. Include the art, TileData and AnimData operations together; include any required Sphere definition/function or other dependencies deliberately. Open Content packs and choose Export pending bundle before saving or clearing that queue.

Share the resulting .uoasset with a short import guide and previews. Its ZIP container retains supported native artwork and animation metadata; recipients import it through Content packs, review destination IDs and retain consecutive frame offsets. A VD is not the distribution format for this artwork. Use Animations > Animated items & effects > Save GIF to prepare a demonstration of saved artwork; distribute the native assets and metadata as well. If you also want to install locally, save the same queue after exporting. Already-saved changes do not remain available to Export pending bundle.

Generated effect functions play a visual. Damage, targeting, cooldowns and combat rules need deliberate server work and testing. C# script generation is separate from the Sphere content-pack remapping workflow: distribute generated .cs files and matching client IDs with their own instructions.

Import a pack

Reconstructed native graphic imports use the same current ownership and replacement review. A portable pack or restored queue does not carry permission to overwrite existing recipient graphics. Inspect the current destinations and give fresh consent for any supported replacement.

Open content pack loads its inventory. Double-click an asset row to set an explicit destination ID in decimal or 0x hex. Review every remap, especially connected equipment IDs and consecutive effect frames. Enable Allow replacement of occupied asset IDs only for an intended replacement.

Review pack import shows destination mappings and linked scripts. Resolve conflicts, then queue/save using the normal workflow. Existing script filenames/definitions are not silently overwritten just because binary replacement was enabled.

Supported remapping and limits

Supported mappings include compatible MUL animation bodies, item/TileData links, known numeric Sphere fields and equipment paperdoll relationships. Animated-item frames must retain their relative consecutive arrangement.

The pack importer does not perform implicit MUL/UOP conversion, arbitrary TileData-layout conversion or universal rewriting of dynamic expressions, macros, custom functions and complex dialog code. Changed body IDs with body-mapping files may require the normal importer. Destination capacities and dependencies still have to match supported formats.

Compiled houses are not supported by generic pack remapping in this version. Use the editable House Builder workflow described in chapter 22. A pack also does not include all saved-world objects or prove that every runtime dependency was captured.

Prepare a recipient-friendly release

Distribute only the intended content and documents. Include a short start-here file explaining the tested Studio version, required client/server formats, dependencies, how to import, what to inspect and how to restore the reviewed operation if needed. Provide previews and original editable source files when you have the rights to distribute them.

For armor, a clean set folder/manifest lets recipients allocate fresh destinations. For animations, keep exported registration metadata with the frames. For houses, include the unlinked design/interchange instructions and any required custom tile art, rather than assuming your multi ID is free elsewhere.

Do not distribute project.json, sessions, account files, world saves, private logs, personal absolute paths or an entire private test-copy folder. Review generated scripts/manifests for dependencies and location assumptions. Include applicable licences for anything you redistribute. Studio's installed licenses folder describes its bundled software dependencies; it does not grant rights to unrelated content.

A release acceptance checklist

- The recipient can follow the instructions without your folder layout or accounts.
- The pack opens or source folder loads on a separate compatible project.
- Destination IDs are reviewed against that project's current data.
- Required scripts and resource entries are included and correctly loaded.
- Both inventory/paperdoll and world appearance are checked where applicable.
- Movement, riding and relevant actions are present for worn or rideable content.
- The retained-backup/recovery route is understood before production installation.
- Version, formats, known limitations and content licences are stated plainly.

29. Troubleshooting and common questions

Project setup says there is no scripts directory

Use version 0.33.13 or newer and choose the actual Server type. For RunUO or ServUO, select the server root or its Scripts directory. For ModernUO, select the checkout or Projects/UOContent; C# content does not need to be moved into a Sphere-style scripts folder. If Auto finds C# but cannot identify the fork, choose RunUO, ServUO or ModernUO explicitly. Check that you selected extracted source or a supported source ZIP containing `.cs` files rather than only compiled server binaries. C# files then open read-only in Script files; empty Sphere definitions are expected for a C# profile.

I only want the artwork, without new scripts

Keep Artwork only selected above the import tabs. This is the default for items, animated items, spell effects and ordinary body animations. Review artwork prepares native client changes without `.scp`/`.cs`, `spheretables.scp` registration or body-remapping changes. You do not need to fill script names, stats or destinations. For raw mount VD data use Import Animation > Body / character animations; the complete Mount workflow includes required riding/server links. Turning script output off does not delete source saved by an earlier import.

The animation library is empty or only shows named creatures

Choose All animations rather than Shard definitions, then choose Body animations or Animated items & effects. Clear Search and choose All sources. Body animations needs complete extracted MUL/index pairs or AnimationFrame UOP archives; animated item/effect sequences need AnimData and compatible item artwork. Use Manage > Edit paths to confirm the active project's actual client folder, then Reload library. You do not need to add a Sphere definition to browse populated client records.

If reading fails or source notices appear, inspect the named file and reason. A source change while reading rejects that result; reload after the files are stable. Optional broken names/mappings can leave physical body previews available while routed lookup or conversion still fails. Keep the original files and use the correct complete client data rather than filling absent actions or frames with guesses.

A raw UOP body previews, but Save entire body as .vd is disabled

The library is showing physical archive groups without AnimationSequence aliases. Its Convert routed body to .vd and Routed body export options commands explicitly use the client layout, which may choose different groups. Alternatively enter the intended logical body ID, choose Source Auto and Load body to inspect the routed body first. Export frame and Save GIF use the displayed raw sequence. A successful preview alone does not establish a supported whole-body VD layout or full movement coverage; read the conversion and coverage report.

An animated item/effect has missing frames or differs from the game

Hover over its Animations library result to inspect the actual frame IDs and source warnings, then double-click it to load the sequence in Animations. Missing frames are reported rather than silently omitted. Preview in Test Scene is optional; Effect frames 0 follows AnimData there. A signed offset can reference artwork before the base ID, and repeated offsets remain repeated frames. Confirm all referenced pictures exist in the effective MUL/UOP source. ArtLegacyMUL.uop takes precedence in the automatic picture reader when present.

These previews do not apply legacy art.def/verdata overrides; their presence is reported. A TileData animated flag that is off can mean the sequence needs explicit effect playback. Preview FPS does not promise the native

automatic-animation or spell timing. Use the matching client and server to check the intended playback rules, and use the normal reviewed import/allocation workflow for any repair.

I cannot see all the panels or buttons

Check the Simple / Advanced switch first. Simple hides the detailed destination, alignment and replacement controls in supported importers and reduces the sidebar. Choose Advanced to reveal them, or press Ctrl+K and search the complete All tools collection to open an editor.

Maximise or resize the window and inspect its scrollbars. Long import forms and the workshop's Advanced settings scroll so their detailed controls remain reachable. Drag the main pane dividers so the script directory and central editor each have space. House settings has a scrollable area; Test Scene has a sidebar with Preview and Room tabs. The current release keeps save-review actions separate from its scrollable details. Check the title-bar version if your installed app still shows an older layout.

My converted VD has only one action or facing

Save sequence as .vd exports the sequence you loaded. It leaves the other slots empty. Choose Convert body folder to .vd for an exported collection, or author and import the additional supported action/direction sequences. Changing the layout does not generate those missing frames. Check movement and riding coverage in the target client before treating a body or wearable as complete.

I keep having to uninstall before upgrading

Use the current Setup directly. It should detect the earlier installation and offer an upgrade/repair in the same location. Save and close Studio first. An older Program Files installation may need Windows elevation using the same account. Check that a shortcut is not launching another old copy elsewhere.

No free block fits, even though I see empty slots

A blank picture can still be Referenced by metadata, an indexed script, a client mapping or overlapping native body records. Read its reason and choose a compatible free range. A Queued slot belongs to another pending operation; inspect Pending changes before trying to reuse it.

If confirmation is requested again, the current proposal or checked files differ from the previous review. Open graphics review details and confirm the exact updated replacement. Earlier consent is deliberately not reused.

The import may need a consecutive range or a whole compatible body block. Individual free cells scattered through an index do not form such a block. Check source/bank, layout, pending reservations and the batch size. Use another supported range or reduce an independent-image batch. New equipment can opt into its specific capacity expansion, but still requires icon and both paperdoll destinations.

Do not use Replace existing to get past an allocation error unless you actually intend to replace those records. A No UOP entry label is not permission to write unsupported UOP body animation.

The app says a file changed after review

The plan was prepared against an earlier revision. Another editor, a previous save or a queued/draft conflict may have changed its dependencies. Keep the current files, reload/review the intended operation and resolve older queue entries. A house update also needs its matching compiled-project link. Do not disable the check by copying stale archives over the target.

My asset looks correct in Studio but unchanged in game

Check the active profile's linked client folder, the game's actual data folder, and Sphere's MulFiles location. Check whether the client uses UOP while you edited MUL, or another override is taking precedence. Confirm the numeric art/body/gump ID used by the script. Save the queue, reload/restart the relevant program and test again. Use Check game files to gather evidence.

The character disappears, slides or faces the wrong way

Inspect the body bank, raw versus Sphere body ID, layout and action/direction. A single image sequence does not create all directions. Check ground origins and compare stand/walk/run. Display directions 5-7 may be mirrored. For a persistent correction, edit/export/import the frames; preview offsets do not modify the saved animation.

Bulk export finished but there is no animation file

Check the VD count and the status. No files exported means only reports were created. The requested ID may be a client alias, map to another bank/body, or use modern UOP archives while the bulk scan checked MUL banks. Open Animations, enter the body ID, choose Source Auto and Load body, then use Save entire body as .vd. Supported UOP ANIMAL bodies produce a standard VD and companion complete ZIP there; unsupported layouts stop with an explanation. Check the coverage report for empty actions or facings before distributing. Chapter 15 covers the full workflow.

Whole-body export cannot identify an Animal that previews correctly

Use version 0.32.2 or later. It reads the same last valid mobtypes.txt definition as the routed viewer, fixing duplicate entries such as body 1729, and supports the additional Animal flag used by body 1638. If the message reports a missing file or body entry, use Edit paths to select the client folder containing the matching mobtypes.txt, then reload the routed body. Another effective body class or an unsupported flag/layout still needs a matching supported conversion; a physical library preview alone does not establish an Animal layout.

My mount is a vehicle, not an animal

Choose Mount > Server script > Mount behaviour > Rideable object when generating the new import. Use the generated inventory-item command. Double-click the item in your backpack to mount; dismount returns the item. Existing animal scripts are not automatically converted. Test ownership and mounting behaviour on the actual shard.

A wearable shows in my backpack but not on the model

An inventory icon is only one asset. Check TileData's layer/animation, the actual equipment VD, male/female paperdoll gumps and their source format. Load appearance and check action coverage. Use Wear in Test Scene for wearing; Show in Test Scene can prepare ground placement. Preserve full paperdoll canvases and provide each armor piece's own assets.

Can I split one complete armor VD into separate pieces?

The set builder does not split a whole-suit VD. Supply separately authored piece animations, individual inventory icons and male/female paperdoll pieces. If the source is intentionally one full-body costume, treat it as one wearable appearance rather than pretending it contains independent equipment layers.

The definition is still referenced outside the removal group

Open the reported script and line, read the use and decide whether it belongs in the removal. Entire animation group can include other definitions that share that animation, but unrelated external code still requires review. Update the reference or keep the shared content. Definition-only removal leaves assets/references for manual handling.

Removal did not delete the .scp file

Inspect the whole file. It may still contain unrelated sections or functions. Use Remove empty .scp file only when appropriate, or Delete script to review intentional whole-file deletion. That does not automatically delete the script's entire transitive asset set. Read REMOVE and KEEP instead of relying on the filename alone.

A script reports a 'charmap' decoding or encoding error

Keep the original bytes and identify the file named in the error. Reload with the current Studio version; supported script reads preserve normal UTF-8/Windows-1252 handling, but not every external file is guaranteed valid. For generated names, shorten them and remove unsupported characters where the client field requires Windows-1252. Do not resave an entire script collection with characters discarded. Include the exact error and a minimal non-private sample when seeking help.

A generated script says the filename or identifier already exists

Choose a unique named definition and filename across the script tree. Reusing an art base through ID inheritance is different from defining that numeric base again. For a linked House Builder project, use its update workflow; for an ordinary existing .scp, edit it deliberately in Sphere scripts. Do not create another conflicting copy simply by changing folders.

Save fails with access denied or a file-lock message

The Save safety check lists unavailable files before a reviewed save creates backups or prepares output. It can also detect another native archive in the selected client folder being used, even when that archive is not part of the edit. Close the client, Sphere and other editors using the reported files, confirm the profile points to a writable working location, then retry. Studio preserves Windows permissions; it does not reset them globally or close programs for you.

An initial failed check has made no partial game-file save. A second check can stop the operation after preparation but before replacement. If the message instead reports an interrupted save or requests recovery, keep its journal and follow Save / recovery centre before retrying. Chapter 4 explains the difference.

Attach sound asks me to save or discard edits

Finish open script drafts before opening the attachment window. If the same script has a queued change, save it or remove it, then reopen Attach sound against the current file. Finish a queued replacement of the selected saved sound first too. This avoids silently combining a new attachment with an older reviewed snapshot.

My attached sound does not play, or plays twice

Test Scene does not run sound events. First use Play in Attach sound to check the selected audio, then verify the actual item event, creature field or function in the target client/server after reloading. Check the selected script section and event, inherited overrides, sound.def mappings and loose audio overrides. An effect function needs to be called; an art ID alone is not a sound trigger.

An item event or function may already contain custom SOUND commands. Studio preserves those and adds its marked command at the start, so both can play. Review the script and deliberately adjust timing or duplicates. Do not delete unrelated event logic.

Sphere will not start after a save

Keep the latest Sphere startup error. Run Item artwork > Check game files and save its report. Compare the affected files and generated script/resource registration with the reviewed operation. Check client/server data paths and compatible formats. Restore the retained linked backup if rollback is appropriate and there have been no intervening changes. Do not repair an unknown problem by mixing unrelated archives and indexes.

My backup refuses to restore

A later edit may have changed one of the operation's files. Recovery stops rather than discarding that newer work. Keep both the backup and current files, identify the intervening changes and plan a merge or deliberate rollback. A whole-file recovery affects all changes inside that archive, not just one visible asset.

My house will not compile or update

Check the required floor, working door and exactly one compatible house sign. Check plot bounds, height, native artwork and matching server housing support. Review headroom/entrance warnings. For an update, use the saved linked project and save queued changes first. If linking a 0.23.0 house, start from its exact original design/name/settings before modifying it.

The updated house in the world still has old fixtures

The compiled definition can change while a placed world house retains its saved door/sign/region state. Redeed and place a fresh instance for structural changes, with appropriate handling of ownership and belongings. House Builder does not migrate occupied world objects automatically.

Does Test Scene prove my spell, pet or house works?

It proves only the aspects you actually inspect in that visual simulation. It does not execute Sphere events or fully reproduce terrain, collisions, equipment sorting or permissions. House Builder's local checks are also approximations. Use the target client/server for gameplay acceptance after visual preparation.

How do I ask for useful support?

Include the Studio version, tool/button used, asset type, selected source format and a clear reproduction sequence. Copy the exact error, state whether changes were queued or saved, and mention any other editors in use. Add the read-only file-check report where relevant. Remove credentials, accounts, world saves and personal paths before posting publicly.

30. Task finder and keyboard reference

Choose the right tool

I want to...	Start here
Find an item or creature definition	Left search and Category / subsection
Inspect linked assets and references	Workspace and Definition
Change a definition or whole script	Sphere scripts; Script directory for full files
Generate Sphere, RunUO, ServUO or ModernUO scripts for existing artwork	All tools > Generate server script
Edit several scripts or replace text	All tools > Sphere script workspace
Inspect an icon, TileData or paperdoll	Item artwork
Find candidate art/body space	Asset spaces or the importer's slot browser
Make a basic item using existing art	All tools > New Sphere item > Use existing artwork
Prepare an external Fiddler import kit	All tools > New Sphere item > Prepare new PNG for UOFiddler
Import floor, wall or roof images	Home or Import artwork > Import Tiles
Import item, furniture or decoration images	Home or Import artwork > Import Statics
Import an animated item	Home or Import artwork > Import Animation > Animated items
Import an effect and optional visual function	Home or Import artwork > Import Animation > Spell / visual effects
Import a monster body	All tools > Creature or monster
Import body animation without a creature script	Home or Import artwork > Import Animation > Body / character animations
Make a mount or rideable object	Home or Import artwork > Import Mounts
Import one wearable or weapon	All tools > Clothing or weapon
Import a complete armor set ZIP or extracted folder	All tools > Complete armor set > Open ZIP or Open set / collection folder
Browse client body animations without a definition or manual ID	Animations > All animations > Body animations
Play or export a body sequence	Animations > double-click a body source; select Action and Direction

I want to...	Start here
Browse, play or export native animated item/effect artwork	Animations > All animations > Animated items & effects > double-click; use Play, Save GIF or Export frame
Convert PNG/BMP frames or a GIF directly to native VD	Home or Import artwork > Animation workshop & VD converter
Convert an exported body folder or ZIP to VD	Animation workshop > Convert body folder to .vd or Convert body ZIP to .vd
Edit frames or reimport a body folder	All tools > Animation workshop & VD converter; Advanced reveals frame editing
Combine rider, mount and equipment	All tools > Equipment & mount preview
Import interface or paperdoll images	Home or Import artwork > Import Gumps
Inspect or replace an existing interface image	Gumps
Design a Sphere dialog	All tools > Sphere gump designer
Edit/test colour tables	Hues
Import, trim or export sound	All tools > Sound workshop
Link a saved sound or WAV to a thing	Asset workspace > Attach sound to this asset...
Play a sound when an effect function runs	Open its script, then All tools > Attach a sound
Preview assets in a room	Test Scene
Build a fixed house and placement deed	All tools > House Builder
Generate a separate wilderness map	All tools > World Generator
Remove a definition and related content	Definition selection > Remove related content
Delete a complete script	Script directory > right-click > Delete script
Export many saved images or VDs	Bulk export
Save all available actions/facings of one body as VD	Animations > Source Auto > Load body > Save entire body as .vd…
Resume an interrupted export	All tools > Resume export

I want to...	Start here
Convert UOP archives to MUL/index files	All tools > MUL / UOP converter > Direction: UOP to MUL
Pack MUL/index files into UOP archives	All tools > MUL / UOP converter > Direction: MUL to UOP
Diagnose file/path mismatches	Item artwork > Check game files
Compare records between clients	All tools > Compare projects
Create a separate development copy	All tools > Create test copy
Share reviewed imports/scripts	All tools > Content packs
Restore drafts or a file backup	Asset workspace > Save / recovery centre

Keyboard and mouse actions

Shortcuts act on the focused control. Click the script editor or scene canvas first; typing in an entry field should not be confused with moving a character.

Context	Action	Result
Main window	Ctrl+K	Search the complete All tools collection
Main window	Ctrl+H	Return Home
Main window	Alt+Left / Alt+Right	Move through view history
Main window	Ctrl+Shift+S	Save All
Main window	Ctrl+P	Open Workspace
Sphere scripts	Ctrl+S	Validate/save the current script context
Script Find	Ctrl+F	Focus Find
Script Find	F3 / Shift+F3	Next / previous match
Find entry	Enter / Shift+Enter	Next / previous match
Find entry	Escape	Return focus to script text
Script workspace	Ctrl+Space	Completion suggestions
Script workspace	F12	Go to definition
Multiple-selection lists	Ctrl/Shift-click	Select multiple rows where supported
Flag picker	Double-click / Space	Toggle selected flag
Item artwork ID	Enter	Inspect the entered art ID

Context	Action	Result
Test Scene canvas	Arrow keys	Local movement
Test Scene canvas	Space	Play/pause
Test Scene canvas	Escape	Return to movement tool
Map Regions canvas	Arrow keys / Shift+arrows	Pan the view; Shift moves faster
Map Regions canvas	Wheel / + / - / Home	Pointer-centred wheel zoom, centre zoom or fit the map
Map Regions canvas	Left-drag in Pan map / middle- or right-drag	Pan the view; middle/right work in every mode
Map Regions canvas	Escape	Cancel an unfinished pan or boundary drag
Map Regions list	Double-click / Enter	Locate the selected region on its facet
House Builder canvas	Middle-drag	Pan view
House Builder canvas	Right-click	Erase hit piece
House Builder canvas	Delete	Delete selected pieces
House Builder canvas	Ctrl+Z / Ctrl+Y	Undo / redo design changes
House walk test	Arrows / WASD	Continuous local movement
House walk test	E / Space	Open/close a nearby door
House walk test	Escape	Stop walking

Generated script destinations

These defaults organise new files; they do not automatically move older scripts. Inspect the destination during review and keep filenames unique. The gump designer asks you to choose a new .scp inside scripts.

Import kind	Default script folder
Monster	scripts/monsters
Mount / rideable object	scripts/mounts
Item	scripts/items
Clothing	scripts/clothing
Weapon	scripts/weapons
Effect	scripts/effects
Armor set piece	scripts/armor
Compiled house and deed	scripts/housing/custom

31. Formats, glossary and version boundaries

Files you will encounter

File/family	Role in Studio
.scp	Sphere definitions, functions, dialogs and resource configuration
art.mul + artidx.mul	Indexed item artwork
ArtLegacyMUL.uop	UOP item artwork
tiledata.mul	Item/land metadata; Studio's editor focuses on supported item records
anim.mul/anim.idx and numbered banks	Supported creature/equipment animation sequences
AnimationFrame UOP family	Modern body-animation viewing/export sources
animdata.mul	Animated-static frame offsets/timing data
gumpart.mul + gumpidx.mul / gump UOP	Interface and paperdoll artwork
hues.mul	Colour-table records
sound.mul + soundidx.mul / sound UOP	Native sound effects
multi.mul + multi.idx / MultiCollection.uop	Compiled fixed structures/houses
.vd	Compatible UOFiddler animation-body transfer
PNG/BMP/GIF	Image and animation-frame source/export formats, according to the tool
PCM WAV	Sound import/export interchange
frames.json / body.json	Animation export registration and structure metadata
.uoworld.json	World Generator seed, settings and painted biome regions
mapN.mul / staticsN.mul / staidxN.mul	Matching native terrain and scenery set from World Generator
.house.json	Editable House Builder document, including an optional destination link
Gump design JSON	Editable visual dialog layout
Blueprint CSV	Supported unhued CentrED tile interchange
manifest.json	Set/collection description for supported armor-folder import
.uoasset	Portable bundle of deliberately queued assets and linked content

File/family	Role in Studio
project.json	Local profiles/settings; not public content

Keep data and index files as matching pairs. Preserve metadata next to exported frame folders. Recognising a filename does not mean every historical/client-specific variant can be rewritten by every tool.

Glossary

Allocation: the existing range of records a format exposes. A compatible free body block can be larger than one visible slot.

Art ID: the number selecting item artwork and its metadata, distinct from a character body or gump ID.

CHARDEF / ITEMDEF: Sphere character and item definition sections. A named definition can inherit from an existing numeric base using supported links.

DEFNAME: a stable script symbol used to refer to content by name.

Gump: UO interface artwork, or informally a complete dialog using that artwork. Studio separates the artwork browser from the dialog designer.

Hue / Partial hue: a palette transformation; partial hue limits colouring to applicable grey pixels. A preview toggle is not necessarily a saved item flag.

Layer: an equipment slot on a character, or an independently composited image in the layered preview. Context determines which meaning applies.

Mapping: a file/relationship directing a logical ID to its actual asset or bank. Changing a raw record does not necessarily change the mapping that selects it.

Multi: a native collection of item components used for a fixed structure such as a house. Functional door/sign objects can be supplied separately by server definitions.

Origin: the registration point used to align frames with their ground/body anchor. Stable origins reduce sliding and jumping between frames.

Paperdoll: the equipment view of a character in the UO interface. Its clothing images are separate from moving world-character art.

Queue / reviewed plan: a snapshot of prepared changes waiting for a game-file save, with dependency and conflict checks.

Recovery journal: the information and originals used to recover a linked save. Keep its complete folder intact.

Session checkpoint: stored editing state; it does not install the reviewed queue into game files.

Static / animated static: item artwork placed in the world; animated statics use frame metadata rather than a creature-body action layout.

TileData: client metadata describing properties such as art name, height, flags and equipment links. Sphere can supply additional or overriding behaviour.

What this release does not promise

The World Generator exports separate new maps. It can plan named destination plots and routes and place selected building stamps; it does not create town populations, guard regions, housing ownership or NPC spawns. Native route checks use a movement approximation and still need receiving-client/server testing.

Version 0.32.0 does not contain a complete Sphere interpreter, full UO game-client emulator, networked CentrED service or general world-map editor. Its visual scenes do not establish server gameplay correctness or execute attached sound events. File-access checks reduce save conflicts but cannot detect every program that has loaded data or prevent all later file opens. The house tool targets fixed compiled houses, not every custom-housing protocol or automatic migration of occupied houses.

Static reference and free-space checks do not resolve every dynamic expression, saved world object or client override. Modern UOP animation does not automatically convert to complete MUL bodies. Ordinary imports do not grow every allocation; the equipment expansion option is a specific reviewed capability. Content packs do not collect every dependency transitively or remap compiled houses.

32. World Generator: create a separate wilderness

Open **All tools > World Generator**, or choose World Generator in Home's Create or convert area. The generator builds a new landscape from the selected client's existing terrain and scenery artwork. You can create a whole supported facet, preview it inside Studio and save its files separately. The linked client and server files are not replaced.

Simple exposes the world preset, map size, landscape, vegetation and refinement controls. Choose Advanced for custom dimensions, water/route targets, destination counts and generation rules. Switching modes reveals or hides controls while retaining the loaded settings. Changing generation values requires a fresh generation before exporting the matching result.

First generated world

1. Choose a World preset: Temperate Mainland, Island Adventures, Frozen Highlands, Dry Frontier or Volcanic Wilderness. The description explains its intended landscape. Presets retain your map dimensions, map number and seed.
2. Choose Small test world for a quick 512 by 512 preview, or the intended full map size. Advanced exposes Width, Height and Map number.
3. Enter a World seed, or keep the supplied starting seed. Choose Generate world.
4. Watch the current phase, elapsed time, completed phases and small live generation pictures. Cancel leaves the last completed world available.
5. Inspect the overview, UO terrain preview and Terrain review. Requested/created counts explain features that could not fit; inspect marked terrain and route problems.
6. Choose Keep this world before trying Another world. Another world changes the seed and generates in one action; settings, painted regions and template choices remain.
7. Refine the world with Reroll chosen part or queued biome painting. Compare worlds shows the completed candidates at the same tile scale.
8. Export new map folder when the generated preview matches the settings. Export remains separate from your linked client and server.

The UO terrain preview uses the linked client's native terrain and planned scenery. The overview's marker identifies the inspected location. These views and the native route checker help inspect a map; they do not run a game server or establish every shard's movement behavior.

Keep and compare worlds

Keep this world retains one complete candidate in memory: settings, regions, templates, native terrain, review and stage pictures. Keeping another candidate replaces that retained candidate. Restore kept returns to its full generated state; changed source artwork requires regeneration before export. Closing World Generator releases these in-memory candidates, so Save setup or Save favourite is the way to retain its configuration for later sessions.

Compare worlds compares the current candidate with the kept world, or with the previous generation's small overview. It includes seeds, settings and requested/created counts. The previous overview is a bounded comparison picture, not another retained full map.

Refine one part

Choose a part and use Reroll chosen part. Each part has its own saved variation number.

Part	What is retained and rebuilt
Land and climate	New land and dependent features; painted regions and template choices remain
Water	Base land shape; new rivers/lakes and affected banks, destinations and approaches
Destinations and paths	Land and waterways; new plot sites and route network
Trees and scenery	Native terrain, water, destination planning and interiors; new procedural decoration
Dungeons and caves	Outdoor planning; new interior floor plans and entrance selection

Dependent features are rebuilt where necessary. For example, changing water can change the dry sites available to roads/buildings. Changing scenery retains the native terrain. Older saved worlds use legacy rules; applying a modern preset deliberately selects current rules.

Building templates and scenery themes

Under **Buildings from your stamps**, choose Auto or an available named destination for the next stamp, then **Add stamp…** and **Generate world**. The setup embeds the saved House Builder `.house-stamp.json` component references instead of its source filename. The stamp must fit a flat, dry plot, use artwork available in the linked client and stay within native height limits. To change an added stamp's target, remove it and add it again. Placement failures are reported. Fixed buildings appear in previews and exported scenery; procedural trees stay clear of reserved plots and required approaches.

Adding a stamp to a legacy setup selects generation rules 4. If its destination count was zero, Studio requests four plots. The next generation uses that updated setup; save the legacy setup first if you want to retain both versions.

Classic UO, Lush, Sparse and Ruined scenery themes use available native artwork. Tree size and vegetation density remain separate choices. A placed building has no automatic ownership, deed, guard-region or NPC behavior; create those deliberately in the server tools.

Explore the world in a separate window

After generating, choose Explore world… above the preview tabs. This opens a separate isometric view of the generated landscape. Resize it or choose Maximise / restore to use more screen space. The window starts at the location selected in the generator. Closing it leaves the generator and its world available.

Control	What it does
Drag the map with any mouse button	Pan across the landscape
Mouse wheel or Zoom	Choose 25%, 50%, 75%, 100%, 150% or 200%
Arrow keys or WASD	Travel in the corresponding screen direction while the map has focus
Shift with movement keys	Move three times as far per step
X, Y and Go	Travel to a tile coordinate inside this world
Click World map	Jump to that location in the whole-world minimap

Control	What it does
Jump to a feature	Visit a volcano, lake, clearing, entrance, interior or reserved black area
Follow entrance / return	After choosing an entrance/interior, inspect the other end of its connection
Scenery	Show or hide trees, foliage, shoreline pieces and interior walls
Tile grid	Show terrain tile edges
Home or Starting location	Return to the initial location at 100% zoom

Click the main map before using movement keys. While typing coordinates, arrow keys remain available for editing the field. The minimap's white outline shows the approximate visible area, and the marker shows the view centre. The status line gives the centre coordinates, terrain height and zoom. At a map edge, empty space beyond the generated world is expected.

Only nearby tiles are drawn. Movement requests replace older unfinished views, so large maps remain navigable. Trees, shoreline pieces and ground selection use the same planned placement as export. The view remains an approximation of client rendering and is useful for inspecting slopes, shores, scenery spacing and routes.

The explorer displays the last generated world. Changing generator controls alone does not change it; Generate world refreshes an open explorer when generation succeeds. Opening different saved settings closes the old explorer. Pan, zoom, grid and scenery controls only affect the display. They do not edit terrain, export files, move a player character or connect to a server.

Size, map number and seed

Control	Meaning
Small test world	512 by 512 tiles for quick experiments
Britannia	7168 by 4096 tiles
Britannia (legacy)	6144 by 4096 tiles
Ilshenar	2304 by 1600 tiles
Malas	2560 by 2048 tiles
Tokuno	1448 by 1448 tiles
Ter Mur	1280 by 4096 tiles
Custom width and height	Multiples of 8, from 64 to 8192 per side, with at most 32 million tiles
Map number	0 through 5; controls the names of the exported map, statics and index files
World seed	A name or number of 1 to 80 characters used to reproduce the result

Map dimensions and map number are separate choices. Selecting a size does not configure your server or change its existing maps. A small test export needs a separate test setup with matching dimensions; do not substitute it for a larger facet without configuring that setup.

New setups use generation rules 4 with independent variation seeds and embedded building templates. Engine 1-3 files select the maintained legacy rules; they retain the earlier landscape rather than silently adopting the new algorithm. Existing exported maps remain unchanged. Exact reproduction also depends on the selected client's artwork and metadata.

When controls change, the status says that the preview still shows the last generated world. Generate world updates it and enables export again. Find area and World Explorer inspect the currently generated snapshot.

The generator version, settings, painted regions and client artwork all matter for exact reproduction. A later generator version or a client with different native art can produce a different result. Keep the saved settings and exported native files together when archiving a finished world.

Follow generation stages

Leave **Keep generation pictures** checked before generating. After completion, choose **Generation stages…** to compare land/climate, mountains/volcanoes, rivers/lakes, paths/biome edits and the final native overview. Connected interiors are shown when present. Previous, Next and the phase selector change only the displayed picture.

Pictures belong to the last completed world, use bounded memory and do not change the seed or map export. Turn the option off to omit them on the next generation. Changed settings still require regeneration before export.

Height, slope and terrain review

After generation, use View above the World overview to switch inspection layers. Click any location to inspect it using the native UO artwork. Legend… explains the overview colours.

View	Meaning
Biomes	The normal landscape and region colours
Height	Dark lowland through light peaks; blue water stays at zero; reserved space stays black
Slope	Flat terrain is dark, smaller height steps are green, ordinary-limit steps are amber, and steeper rock faces are purple
Shoreline / blends	Cyan shores, purple corners with three or more materials, red tiles without matching shoreline artwork in this client
Native movement	Reachable ground, isolated walking surfaces and required locations using the client's TileData
Transition matches	Exact, approximate and missing terrain-brush matches using available native artwork

Terrain review checks every native tile. Features / suggestions shows requested and created features, including destinations, paths and interiors. Check details… opens the complete report. Inspect: Coast / Entrance / Steep ground / Unreachable provides quick jumps, and the table lists up to 24 spread-out examples per inspection category. Double-click a row, or choose Inspect selected location, to open that coordinate in UO terrain preview.

Terrain junctions and steps at the allowed limit are places to inspect. Native movement checks approximate one walking surface, using real land/static flags, surface height, headroom and cardinal steps. They check the start, named plots and outdoor entrances separately from interior landings/returns. Safe repairs can clear procedural blockers or add bounded bridges when suitable native bridge artwork is available; fixed building components stay intact. This does not run the game's movement rules, NPC pathfinding or server regions. Counts describe the last

generated world; changing controls still requires Generate world before export. Inspection images and sample coordinates are included with exported maps.

Terrain restrictions and logical regions

Mountains setting	Maximum height	Ordinary ground step	Classic UO rock-face step
Gentle	28	2	8
Rolling	52	3	12
Rugged	80	4	16

Mountain edges offers Classic UO or Soft slopes. New worlds use Classic UO: rock interiors rise into craggy faces, with uneven capped ridges and graded margins. Soft slopes retains the ordinary-ground limit everywhere; older projects without this setting open with Soft slopes. Every horizontal and vertical terrain edge is checked before export. Larger steps are allowed only on edges touching rock. Water remains level at zero. A result exceeding these limits cannot be exported.

Raised terrain uses the client's square terrain textures when texidx.mul and texmaps.mul are available, including TexTerr.def mappings. Flat ground retains its diamond artwork. Missing optional textures fall back to that artwork. This improves mountain faces and their grass transitions; the preview remains an approximation of native client lighting.

Climate connects temperature with latitude and elevation. Dry regions favour warmer land, and snowy regions favour cold or high areas. Coastlines receive sand shelves. Snow receives earth margins, and craters receive rock margins. Ground brushes select matching native transition tiles between supported terrain pairs.

Sea % is a target from 25 to 80, and Forest % is a target from 0 to 85. Actual coverage changes when islands, coasts, mountain cores, intermediate ground and protected features are applied. These controls are not promises that exactly that percentage of the final tiles will have one appearance.

Rivers follow a drainage route toward the sea. The generator cuts level water channels and grades their banks instead of laying water uphill across the original slope. Lakes connect into the drainage network. Water surfaces use height zero. Volcanoes need enough surrounding land for their rocky slopes; a small, cold or fragmented island world may have fewer volcanoes than requested.

Rivers, lakes, volcanoes and paths

Rivers accept a target of 0 to 24, lakes 0 to 12, volcanoes 0 to 8, and path links 0 to 16. The result summary gives the actual number successfully placed. A requested feature is skipped when there is no suitable location or route. Try another seed, a larger landmass or less sea when a feature does not fit.

New worlds reserve named Start, Village, Harbour, Ruins and Camp plots on connected dry ground. Their required routes remain even when optional path links are zero. Routes favour gentler approaches and protect plots from procedural trees. House Builder templates can supply fixed buildings, and native review can add short bridges using verified available artwork. Requested destinations may be skipped if they cannot fit; the report explains the result. River and path placement creates map data, not population or town scripts in Sphere.

Maintained legacy rules 3 consider every earlier clearing when seeking a reachable connection and avoid diagonal gaps between blocked shore tiles. Engine-1 and engine-2 settings open using those legacy rules. Preserve an older map's native export if you need its original historical layout.

Reserved black area, dungeons and caves

Under Dungeons and caves, Space for dungeons/caves reserves the right edge of the map for interiors and unused space. Small uses approximately one eighth of the width, Medium one quarter, and Large three eighths. At least 64 tiles remain for the outdoor region. New worlds start with Medium, two dungeons and two caves; older saved projects retain their original outdoor-only settings. Sea % applies to the outdoor region. A reserved world needs a width of at least 72 tiles.

Dungeons and Natural caves accept targets from 0 to 12 each. Interior size selects Small (48 tiles), Medium (80) or Large (128) planning areas. Stone dungeons have connected chambers and corridors. Natural caves have rounded chambers and wider passages. Native walls surround the floor, and black space separates the layouts. Interior floors are level at height zero.

Each placed interior receives a dry entrance on the outdoor map, preferably at a mountain edge. Natural caves use native black cave-mouth artwork with rough rock edges and a clear dirt approach. Their entrance trigger is immediately in front of the blocking artwork, linking to the separate interior rather than a walkable tunnel through the outdoor mountain. Stone dungeons retain an open entrance chamber and central trigger. Its inside return trigger is two tiles north of the arrival point. Landing points are separated from triggers so arriving does not immediately send the player back. Entrance platforms grade into the surrounding terrain and keep water level.

Requested counts are targets. When the black area is too narrow, the selected layouts do not fit, or there is insufficient dry land, the status message reports the actual number. Increase the map or reserved area, choose smaller interiors, lower the counts, reduce sea coverage or try another seed. Unused space stays black. Biome painting cannot replace it. Set both counts to zero to reserve an empty area for later map editing; also set Space for dungeons/caves to None for a completely outdoor map.

In the separate explorer, choose a named entrance or interior in Jump to a feature. Follow entrance / return switches to its other end. Interior jumps choose 50% zoom. Use the wheel to see the whole layout, and switch Scenery off to inspect floor routes without walls. This is an in-app camera preview of the connection, not a running server.

Trees, foliage and saved biome regions

Tree density and Foliage each offer None, Sparse, Medium and Dense. Tree size offers Small, Medium, Large and Mixed. Size and density are independent: Medium trees with Sparse density gives occasional medium trees. Complete tree groups keep the correct canopy with their trunk; separate fragments are not scattered as if they were whole trees.

Vegetation varies with the ground region. Trees and plants are kept away from paths, shorelines, water, lava and mountain cores. Increasing foliage changes scenery density without raising terrain or changing the cliff limits.

Choose a biome under Paint biome regions, set Brush radius in tiles, then click the overview. The radius can be 4 to 2048 tiles; the brush outline shows its extent. Painting queues regions without regenerating on every click. Undo and Redo change the queued strokes; Apply regions builds the updated map. Inspect returns to location selection. Painted regions persist across seed changes. Resizing an already generated world scales saved brush positions. Water, coasts, paths and crater cores retain their existing protection.

A painted region changes the surface biome, not the elevation. Water, shoreline shelves, paths and crater cores remain protected. Regions do not lock every underlying terrain tile, river or mountain in place. The generator applies its normal intermediate ground around painted regions, so an abrupt snow-to-desert brush stroke still receives a buffer.

World setups contains Open setup…, Save setup…, Save favourite… and Favourites / recent…. Save setup writes a .uoworld.json containing the controls, templates and painted regions. Open setup reloads it and generates the preview. Saving a setup is separate from exporting native files. If you change a control after

generating, Export asks you to generate the updated preview first, ensuring the exported result matches what you have reviewed.

Export contents and checks

Choose Export new map folder and select an existing folder outside your linked client and server. Studio checks required source files for open handles, destination write access, free disk space and terrain limits. It writes into a temporary folder, verifies native block sizes and scenery indexes, then publishes one uniquely named result folder. Cancellation removes that operation's temporary output. Existing files in the selected output directory are not overwritten.

Exported file	Purpose
mapN.mul	Native terrain tile IDs and signed heights
staticsN.mul	Trees, foliage, shore pieces and interior walls
staidxN.mul	Native index locating each scenery block
overview.png	Complete world overview
terrain-height.png, terrain-slope.png	Height and slope inspection views
terrain-shoreline-review.png	Shoreline, mixed-material and missing-artwork markers
terrain-native-movement.png	Native walking-surface and required-location diagnostics
terrain-transition-review.png	Native terrain-brush coverage diagnostics
heightmap.bmp	Grayscale height plus 128, compatible with CentrED height import
world.uoworld.json	Reproducible seed, controls and saved biome strokes
world-report.json	Dimensions, height checks, feature counts, scenery totals and terrain-review sample coordinates
dungeon-links.json	Entrance, exit and landing coordinates when interiors exist
server-scripts/maps/generated	Companion Sphere entrance-link script when interiors exist
READ-ME.txt	File-set and separate testing instructions
SHA256SUMS.txt	Checksums for the completed output files

Keep the terrain, statics and index together. A client requiring mapLegacyMUL.uop can use All tools > MUL / UOP converter with Direction: MUL to UOP on the generated mapN.mul. Statics and their index remain separate. The converter uses directly readable terrain chunks and the native internal entry names for alternate map files.

Activate entrance links in a separate test server

When interiors exist, keep dungeon-links.json and the companion server-scripts/maps/generated script with the exported map. The script uses Sphere's [Teleporters] section for entrance and return coordinates. Review its map number and positions, copy it to scripts/maps/generated in a separate test server, and add its path to that server's spheretables.scp include list. Load the matching map, statics and index with the same facet dimensions. Map files alone do not activate these links.

Walk through every entrance and return point in that separate setup. Check wall corners, collision, the clear approach and landing positions using your own client/server rules. Studio does not install the script or change live files. No creatures, loot, lighting regions or server behaviour are supplied by the dungeon layout.

Further editing and deployment

For further editing in CentrED, create a separate workspace containing the generated native file set and your own matching client artwork. Use the exact tile dimensions from the report. CentrED's block dimensions are the tile dimensions divided by eight. The exported height image uses the height-plus-128 convention expected by its grayscale height importer.

Inspect coastline corners, routes, mountain approaches and biome junctions before deploying a world. Stock single-tile brushes cannot represent every three-material junction; the report counts these locations for final editor polish. The preview uses native artwork but is not a complete client renderer or movement simulator.

The world-report.json export also records generator_version and unmatched_coast_tiles. A nonzero unmatched count means the chosen client lacks matching shore artwork for some corners; the completion status calls this out. Inspect the shores before using the map. The report also counts three-material junctions that may need final map-editor polish.

World Generator creates outdoor terrain, optional interiors and companion entrance-link scripts. It does not create NPC spawns, towns, server regions, housing deeds, player start locations or a new server configuration. Existing map patches, regions and saved object positions belong to the old world and need separate review. Test matching client and server copies before replacing any live map; exporting from Studio performs no deployment.

Setup favourites and recent files

Save favourite stores a named setup in Studio's separate private world library. Favourites / recent opens saved choices and recently opened setup files. Large embedded templates may require Save setup instead. An unreadable or externally changed library is protected from silent replacement; an explicit reset preserves a copy first.

33. Map Regions: boundaries and server rules

Open existing regions

Choose Map Regions in the sidebar, or All tools > Worlds > Map Regions. This editor targets Sphere AREADEF and ROOMDEF scripts. RunUO, ServUO and ModernUO region formats are not edited by this release.

Choose the map ID in **Map**; the left list shows regions on that facet. Search by name, identifier or source filename. A dot marks a changed draft. Click a name or map boundary to select it. Double-click a listed region, or select it and press Enter, to locate its boundaries on the selected facet. Overlapping map boundaries select the smallest matching region; use the list for an enclosing region. Find selected fits the region's rectangles in the preview.

The editor reads regions from the project's script collection, including files which may not be active in spheretables.scp. It does not run the server or resolve every custom event. Region ordering and overlapping rules still follow the shard's Sphere configuration.

Load and navigate a terrain map

When it opens, the editor attempts to read the selected facet's mapN.mul or mapNLegacyMUL.uop from the linked client; the default map ID is 0. Load terrain lets you choose a different MUL or legacy map UOP, including a separately generated world. Select the corresponding map ID. Standard facet sizes are detected and loaded automatically, including 6144 x 4096 and 7168 x 4096 Britannia maps. The loaded dimensions appear above the preview.

Custom maps require their actual width and height in tiles, divisible by eight. UOP files do not store width and height separately, so Studio recognises standard sizes from the terrain length. Cancelling the dimensions window keeps the current preview and scale. Version 0.30.1 fixes the incorrect dimension error for standard legacy UOP archives, including their optional trailing padding block; you do not need to convert the UOP to MUL first.

A sibling radarcol.mul supplies terrain colours. Without it, the map uses a labelled grayscale elevation preview. This is a terrain overview: it does not render buildings, statics or live characters. Files are opened for reading, and loading a map does not write client files. Changing the map ID clears the background so boundaries are not drawn over the previous facet by mistake.

Choose **Pan map** to drag with the left mouse button. Middle- or right-drag pans in every mode, including while Draw boundary or Move boundary is selected. Panning moves the view without changing the region's rectangles. Press **Escape** to cancel an unfinished pan or boundary drag.

With the map canvas focused, use the **arrow keys** to pan; hold **Shift** for a larger step. Click the canvas before using these keys. Arrows in the search box, coordinate fields or region list retain their normal editing and list-navigation behavior.

Use the mouse wheel to zoom around the pointer, or use **+** and **-** to zoom at the view centre. **Fit map**, or **Home** while the canvas is focused, shows the complete facet; **Find selected** fits the region. On the Boundaries tab, selecting a rectangle loads its coordinates and **Fit boundary** fits that one rectangle. Without terrain, edit boundaries on the coordinate grid. The status line shows the pointer coordinates.

Changing **Map** switches the facet, filters the region list and clears an unrelated terrain background. Loading a new terrain map replaces its preview only after a successful read; superseded loads cannot replace the newer map. These navigation actions leave terrain and server scripts unchanged.

Adjust, add or remove boundaries

1. Select an existing region, or choose Add region for a new AREADEF with a unique identifier.
2. Enter the display name in Region details.
3. Open the Boundaries tab, choose Draw boundary and drag across the map. The first draw replaces a new region's placeholder; further draws add rectangles to the same region.
4. Select a rectangle in Boundary rectangles. Edit X1, Y1, X2 and Y2, then choose Update boundary. Coordinates are decimal; X2 and Y2 are the first tiles outside the boundary.
5. Choose Move boundary and drag to move the selected rectangle without changing its size. Movement is constrained to the displayed map.
6. Remove boundary removes one rectangle. A new or edited region needs at least one rectangle; Remove region removes the whole region from the draft.

A region can contain up to 256 rectangles, all on one map. Irregular areas are represented by several rectangles. Names and boundaries must pass validation. Regions with expressions rather than literal boundaries are marked for editing in the script editor instead.

Selecting another boundary row or removing one first applies valid current coordinate edits to the draft. Invalid coordinates are reported before changing rows. The rectangle list scrolls with the wheel over it. A click or drag with no nonempty rectangle does not add a boundary.

Apply to draft records the current details locally; selecting another region and opening Review changes also apply valid details. Selecting another region keeps the intended sidebar row after applying pending name or rule edits. Undo restores up to 50 earlier draft states. Closing an edited draft asks whether to discard it. Unreviewed region drafts are not included in session checkpoints, so review and queue work you want to keep between sessions.

Choose the rules

Open the Rules tab in Region details.

Control	Sphere behaviour
Guarded	Enables the guarded region flag; custom guard events and configuration still apply
Safe from harm	Marks the region safe; this takes precedence over guarded behaviour
Allow spells	Clears the global antimagic restriction when enabled
Allow harmful spells	Controls the harmful-spell antimagic restriction
Allow recall / mark into region	Controls the recall-in / mark restriction
Allow recall out	Controls the recall-out restriction
Allow gate travel / Allow teleport	Controls the separate gate and teleport restrictions
Allow PvP / Allow building	Controls player combat and building restrictions
Underground / no weather	Marks the region underground
Announce entry / Instant logout	Sets the entry-announcement or instant-logout flags

Control	Sphere behaviour
Prevent item decay / Arena rules	Sets the corresponding Sphere region flags

Checked means the named rule is enabled. Disabling Allow spells overrides the more specific travel permissions; recall restrictions can also affect Mark and Gate in Sphere. These are region flags, not an arbitrary individual spell blacklist. Custom triggers and server settings can impose further restrictions.

Recognised numeric and standard flag expressions populate these controls. Unknown flag bits are retained when known rules change. A custom expression that Studio cannot resolve makes the rule controls read-only; its original text is retained while supported name/boundary changes remain available. Edit such flags in Sphere scripts when you know the shard-specific expression.

Review, queue and save

Choose Review changes. Save or discard any open script edits first. The review shows the exact script differences, new files and resource-list changes. Original unrelated sections, custom tags, events and the order of retained regions are preserved; existing script encoding and newline format are retained.

Version 0.33.9 preserves exactly unchanged loaded region definitions before checking edits. An unrelated unsupported boundary or legacy definition does not block a valid region's review, and its original script text remains untouched. Leaving an unchanged details form also retains its original values; pending coordinate edits still have to be valid. Edited and new regions must pass the normal boundary and property checks. If validation fails, the error identifies the affected region and source file so you can find the draft that needs correction. Use the script editor for unsupported definitions; keeping their text does not repair or simulate their server behaviour.

New regions go in a uniquely named file under scripts/regions and are registered in spheretables.scp. A new region contains a name, centre point, boundaries and chosen flags. It does not automatically copy custom resource events, spawn rules, guards or scripts from another region. Configure these through the script editor where the shard requires them.

The backup option starts enabled. Queue changes adds the reviewed operation to Pending changes; Save All writes the queued work. Saving immediately uses the same access checks and recovery journal. A changed script collection stops the reviewed operation and requires a fresh region review.

Stop Sphere before saving, then restart it. Keep a covering world region; without one, parts of the map can become unusable. Region removal leaves terrain, objects and saves intact. Verify spell, travel, guard and overlap behaviour on your shard; the overview does not simulate server rules.